
Swing Check (Flap) Valves

'Swing check' valves is the preferred description for non-return valves consisting of a hinged disc, although they are also commonly called flap valves because of their geometric action. Their mode of working is obvious. With flow in one direction the disc hinges upwards to permit flow through the valve. With reverse flow the disc is held closed. Equally, spring pressure or mass effect normally holds the disc closed in the absence of flow. In some cases closure is also assisted by the use of a weighted lever.

Small-size swing check valves for low-pressure services may use an elastomeric disc with a square end clamped in position, eliminating any need for a separate hinge. A back-up plate is added to assist closure and also provide rigidity to the unsupported area of elastomer when the valve is closed. Without this the disc would tend to collapse and extrude through the port. Larger versions of flexible (elastomeric) flap valves are made in sizes up to 1500 mm (60 in).

Larger swing check valves are more usually made with discs or flaps of metal or composite materials, hinged at the top and sealing on a metal seal. The sealing surface is inclined at a small angle to the vertical to assist opening, provide more positive sealing under back pressure and reduce shock when closing under high pressure. A typical valve design is shown in Figure 1. In

1. Body in ductile iron FGS 500-7 epoxy coating 250 microns inside/outside
2. Cover in ductile iron FGS 500-7 epoxy coating 250 microns inside/outside.
3. Bolts in dip galvanised steel.
4. Closing disc: DN 65 mm to 150 mm: ductile iron FGS 500-7
DN 200 mm: steel.
5. Drain plug: dip galvanised steel.
6. Disc coating: NBR (Nitrile).
7. Gasket: NBR (Nitrile).
8. Disc shaft: brass BS.2874 NBR coated.

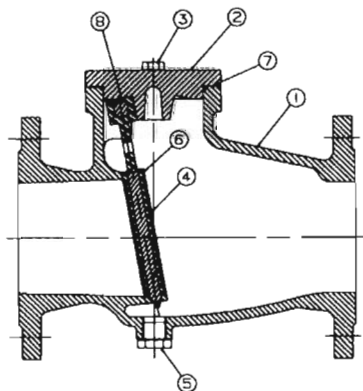


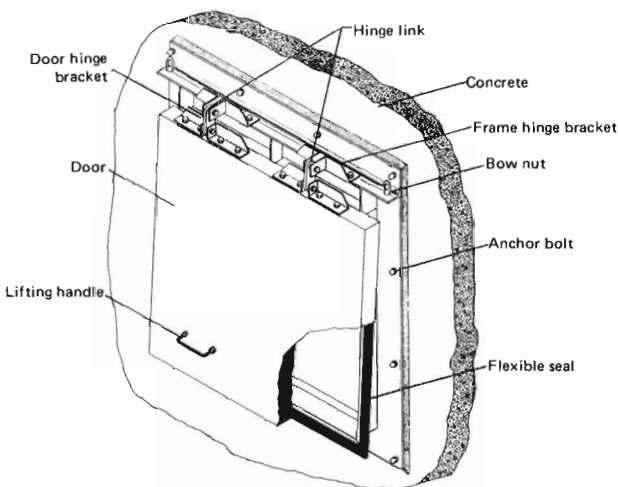
Figure 1. Typical standard swing check valve.

larger valves the disc or flap may be double hung. Flap shapes are normally (but not necessarily) circular.

Swing check valves present relatively high resistance to flow in the open position as well as creating turbulence, because the flap 'floats' in the fluid stream. They may also tend to chatter in systems having frequent flow reversals. Swing check valves are normally used in horizontal pipelines. They can also be used in sectional pipelines with upward flow. They are not suitable for use in systems with pulsating flows. However, weight added to the flap or disc



Standard series swing check valve.



Large flexible-flap valve.

(or spring-loading) can control the opening pressure as well as assisting closure under back pressure.

Commonly the body shape incorporates a ‘dead’ volume in which fluid can be trapped downstream of the flap once the valve is closed. If necessary, a drain can be incorporated at this point, e.g. when it is desirable to drain a system completely on shut-down.

The double-plate check valve shown in Figure 2 is particularly suitable for pumping and general industry applications such as water treatment, irrigation, general circuits and industrial processes.

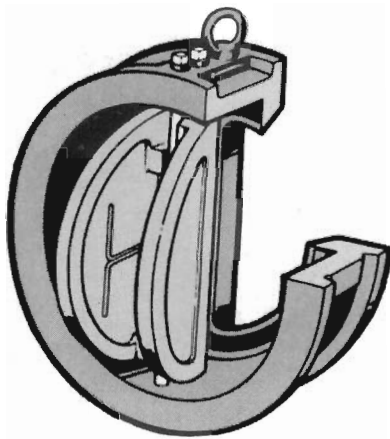


Figure 2. Double-plate swing check valve.

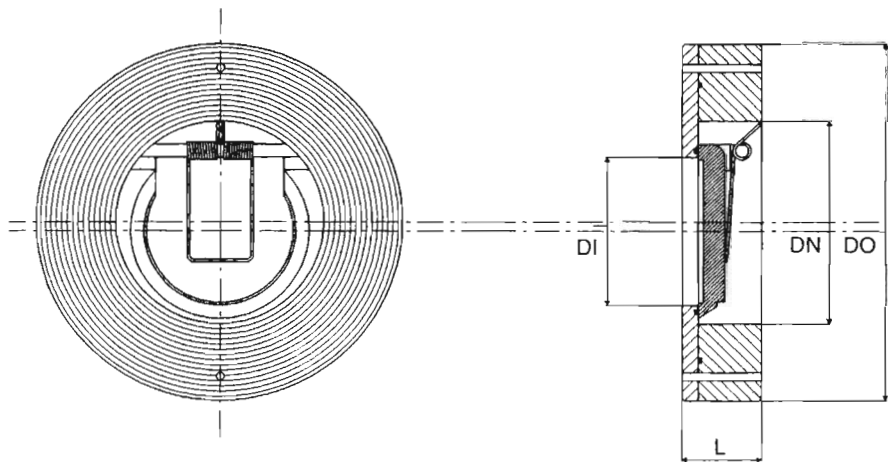


Figure 3. Sprung swing check valve.

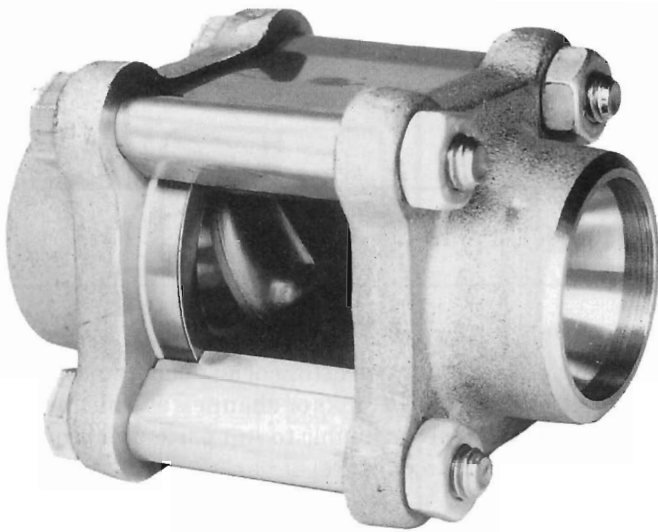


Figure 4. Three-piece construction swing check valve.

The valve is mounted between flanges and is suited to installations where space is limited. The swing check valve shown in Figure 3 has only one moving part and is of the sprung type. It has a tight shut-off and is capable of working in temperatures up to 600°C and pressures up to 100 bar. The spring mechanism is usually made from Inconel-X, titanium or Hastelloy.

Swing check (flap) valves are also produced in a three-piece construction similar to that used in ball valve design (Figure 4). The straight-through flow makes it particularly suitable for hydrocarbon and chemical process lines. For maintenance purposes, the valve body swings outward to allow access to the check and seats.

Materials

Body materials used for swing check valves include cast iron, bronze, cast steel, forged steel, stainless steel, high-duty alloys and also plastics. Valve discs may be in similar materials or composites.

Applications

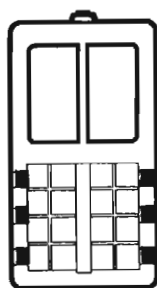
Main application areas for swing check valves are the water industry, including pumping and water treatment, irrigation, and the petrochemical industry including hydrocarbons and industrial processes. Other applications can be found in air conditioning and general industry.

Penstocks

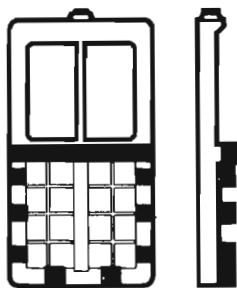
A penstock is a single-faced valve consisting of an open frame and a door. This form of valve is normally located in tanks or channels as a means of controlling flow into a pipe. Many types are available to suit particular requirements and operating conditions. The main ones are:

- (i) Penstocks for operating against pressure, i.e. pressure forcing the door onto the frame.
- (ii) Penstocks for operating against off-seating pressure, i.e. pressure forcing the door away from the frame.
- (iii) Penstocks designed to accommodate both seating and off-seating pressures.

Seating pressure can be accommodated by the use of side wedges only. Off-seating pressure requires the use of bottom wedges or top and bottom wedges (see Figure 1). Penstocks subject to seating pressure normally seal tighter than penstocks for off-seating pressure duties. Both types can be made virtually drop-tight with correct installation, distortion of the gate frame at



Side Wedges



Side, Bottom and Top Wedges

Figure 1. Penstocks showing the location of wedges.

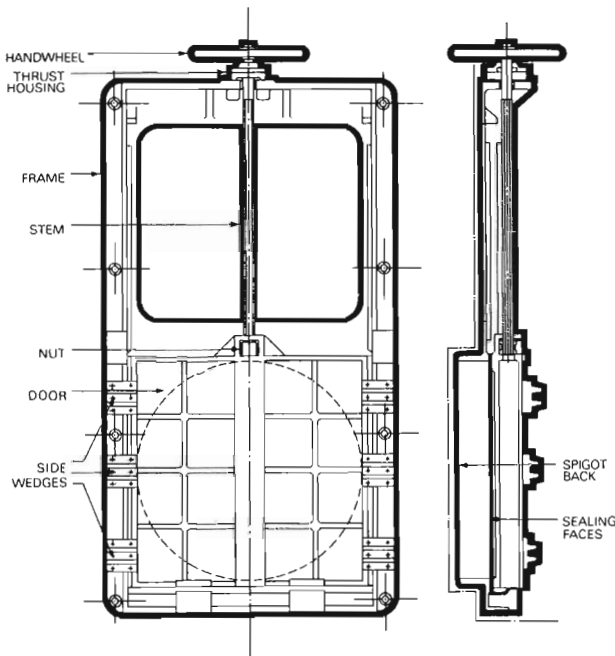


Figure 2. Typical penstock.

the time of installation being the determining factor as far as leakage is concerned.

Penstock frames may be circular or rectangular. In the latter case a preferred proportion of width to depth is 2:3 for vertical form, and 4:3 for horizontal form. Frames and gates are commonly made of cast iron, although plastic materials (usually reinforced steel) are also used for gates operating in aluminium, stainless steel or epoxy-coated frames for corrosive applications. Frames may be for channel or wall-mounted application (Figure 3). Sealing faces in suitable materials are embedded into both the frame and door surfaces.

Handwheel operation of the gate is normal, using a rising stem supported by a suitable headstock or bracket. The advantage of a rising stem is that the screw thread at the bottom of the stem is not usually immersed and is readily accessible for lubrication. A non-rising stem eliminates the need for a headstock and merely rotates through a nut in the penstock door (see Figure 2). The threaded portion at the bottom is then usually immersed in the product being handled. Different systems for raising and lowering the gate may be employed on modulating penstocks used for flow-control purposes (Figures 4 and 5).

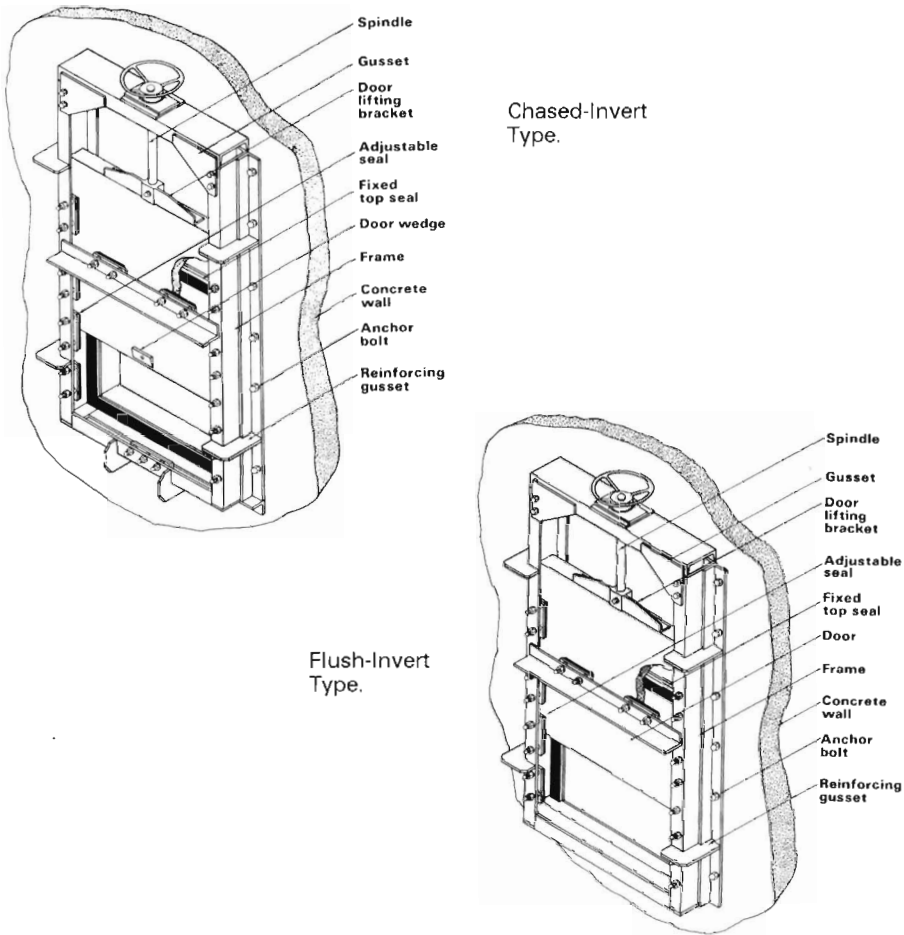


Figure 3. Wall-mounted penstocks.

Discharge through penstocks

A penstock when opened represents a partial obstruction in an open channel over which liquid accelerates with a free liquid surface. Its performance is thus essentially similar to that of a weir where flow rate Q is proportional to width and velocity head. Working formulae are:

$$\begin{aligned} Q \text{ (Imp gal/min)} &= 0.13 \times \text{area (ft}^2\text{)} \times \sqrt{2 g \times \text{head (ft)}} \\ Q \text{ (US gal/min)} &= 0.158 \times \text{area (ft}^2\text{)} \times \sqrt{2 g \times \text{head (ft)}} \\ Q \text{ (m}^3\text{/s)} &= 0.7 \times \text{area (m}^2\text{)} \times \sqrt{2 g \times \text{head (m)}} \end{aligned}$$

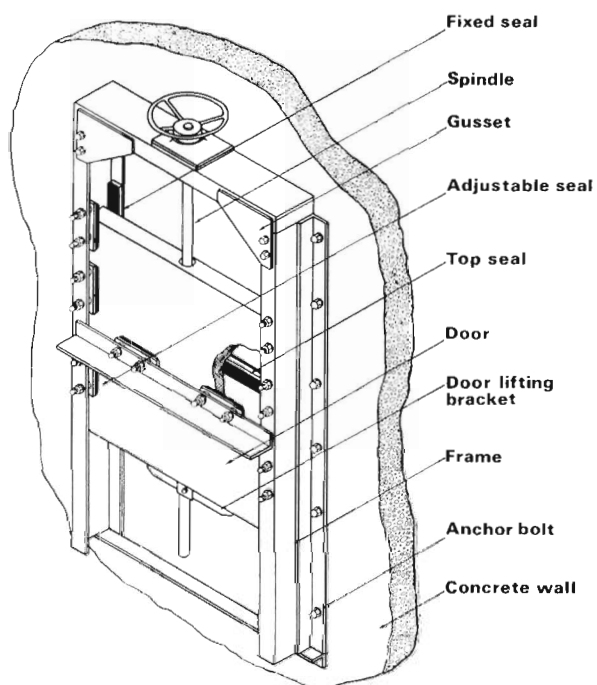


Figure 4. Weir penstock.

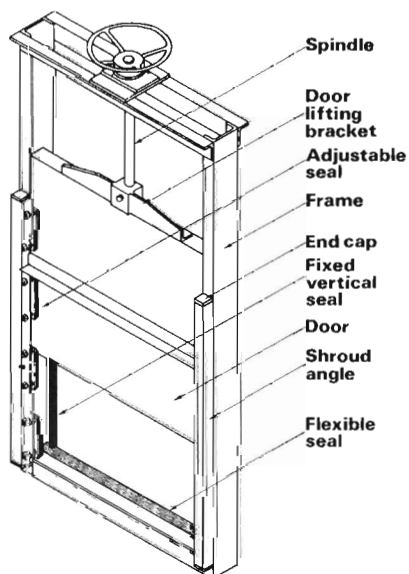


Figure 5. Channel penstock.

Miscellaneous Valves

Where processes or storage applications require continuous safety, the three-way changeover valve allows the user to cross over from one relief-valve system to another. The design permits two valves on a single riser. This arrangement protects the system with one active valve while the other valve is either in reserve or being serviced. Construction details are shown in Figure 1.

The valve works with many general and specialised applications and can incorporate an interlock system that allows either valve to be discharged into

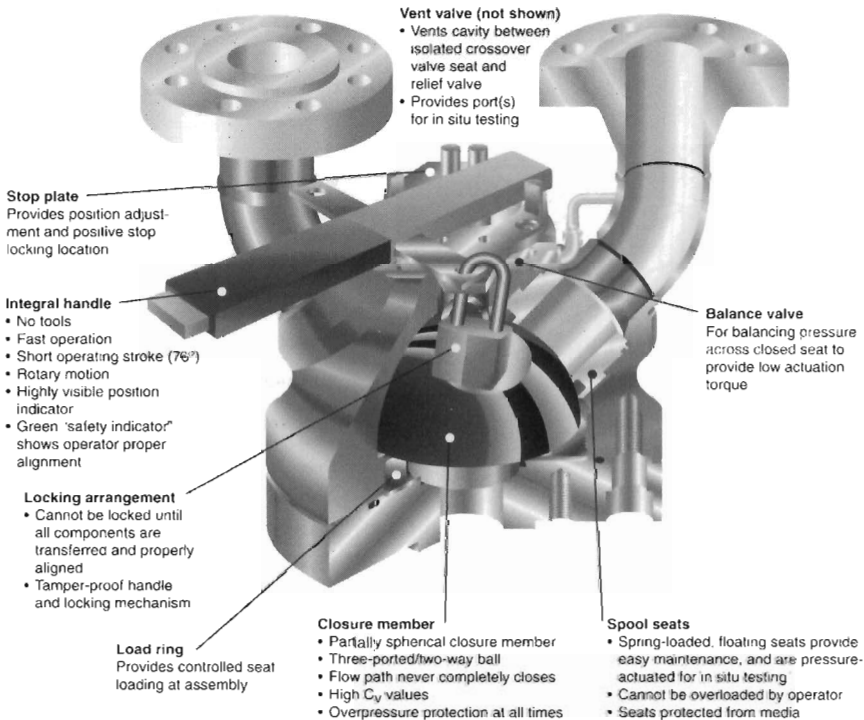


Figure 1. Three-way changeover valve.



Quick cross three-way changeover valve.

a single header. One valve mounts on the riser, the other on the outlets of the relief valves. The valves are operated simultaneously through a simple linkage. This type of valve can also be adapted for remote actuation, still maintaining the single movement rotary change-over action.

Typical areas where the valve is used include chemical plants, fertilizer plants, offshore platforms, refineries, pulp and paper mills, gas distribution systems, toxic service, environmental protection, chlorine storage tanks, refrigeration, dual filtration and process systems.

Pneumatic piston controlled on-off valve

Typically, this type of valve is best suited for controlling the flow of fluids, gases, steam and other substances apart from explosive substances (Figure 2).

The valve is equipped with a position indicator. Generally the material of construction is either AISI 316, bronze or brass. Although the valve can be fitted in almost any position, it must be fitted so that the direction of flow is opposite to the plunger-closing position, otherwise water hammer can result.

Tunnel diverter valve

This valve is designed for use in pressure or vacuum systems to divert or converge pellets: granules, fine powders or abrasive materials. Two types of seals are



Figure 2. Piston-controlled on-off valve.

used: O-ring seals for applications conveying pellets and granular materials, and air-assisted seals for conveying powdered or abrasive materials where three air inlets introduce compressed air into the diverter-valve housing.

An unusual design of check valve is shown in Figure 3. This features a lipped elastomeric membrane as the working element, offering virtually unrestricted flow in the open position with a capability of passing suspended solids up to the full bore diameter. The membrane itself is held open to a full circular form by the flow, the circumference of the membrane in this condition being D . Loss of head is thus minimal (e.g. directly comparable with that of a swing check valve).

With reverse flow, the membrane assumes a closed position with the lips in mating contact (Figure 4), i.e. the natural 'unloaded' form of the membrane. Closure is further assisted and maintained by the reverse flow impinging on the sides of the now wedge-shaped membrane. The length of seal in this case is $\pi D/2$, or substantially half that of a conventional check valve (i.e. the potential leakage path is reduced by half).

The membrane itself is not subject to elastic deformation, merely flexure, and thus has a long life, particularly if the fluid does not contain abrasive solids in suspension. When servicing is required, replacement of the membrane



Figure 3. Membrane check valve.

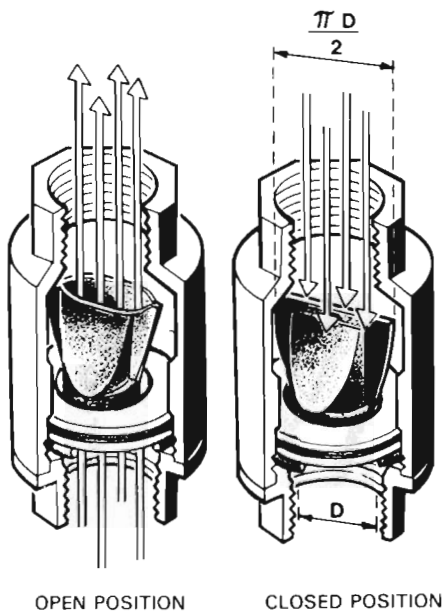


Figure 4.

is a simple operation. No other servicing is necessary. The membrane elastomer is selected according to the product to be handled.

Because of the elastic nature of the closure, this type of check valve cannot water hammer and is also noiseless in that it has no hinge or spring which can be excited into vibration in either the open or closed position.

Currently this valve is made in threaded form for water pipes up to 50 mm (2 in) diameter and flanged for water pipes from 50 to 400 mm (2 to 16 in) diameter. Valve sizes up to 125 mm (5 in) have a single passage. Larger valves have several passages, each with its individual membrane (Figure 5). This solution of dividing the total flow into partial flows with smaller flow rates which never open or close strictly eliminates water hammer in these larger valve sizes with higher flow rates.

Eccentric valves

The description 'eccentric valve' is applied to plug valves having an eccentric motion against a resilient facing, i.e. as the eccentric plug rotates 90° from open to closed, it moves into a raised eccentric seat. The complete action can be followed from Figure 6.

In the open position, the segmented plug is out of the flow path. Flow is straight-through, flow capacity is high. As the plug closes, it moves towards the seat without scraping the seat or body walls so there is no plug binding or wear. Flow is still straight-through.

In the closed position, the plug makes contact with the seat. When furnished with a resilient facing, the plug is pressed firmly into the seat for dead-tight shut-off. The eccentric plug and seat provide lasting shut-off because the plug continues to be pressed against the seat until firm contact is made.

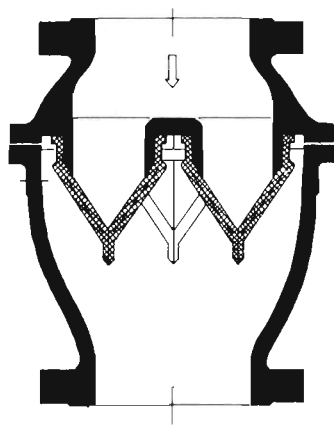


Figure 5.

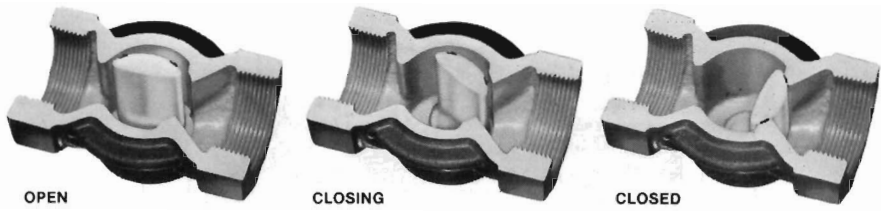


Figure 6. Eccentric valve: complete action.

Throttling characteristics of a valve of this type are generally excellent (Figure 7), and shut-off in the closed position is positive with air and gases as well as liquids.

The valve shown in Figures 8 and 9 has been specifically intended for applications such as tanker truck loading, portable tanks, intermediate bulk containers (IBCs) bulk transfer stations, and agricultural and aviation duties.

The unit is designed to avoid accidental spills and cannot be disconnected in the 'open' position. A handle prevents this and poppets automatically stop the flow from both directions when the unit is disconnected. Maximum working pressure is 150 lb/in². The valve must not be opened or closed when the pressure is in excess of 60 lb/in².

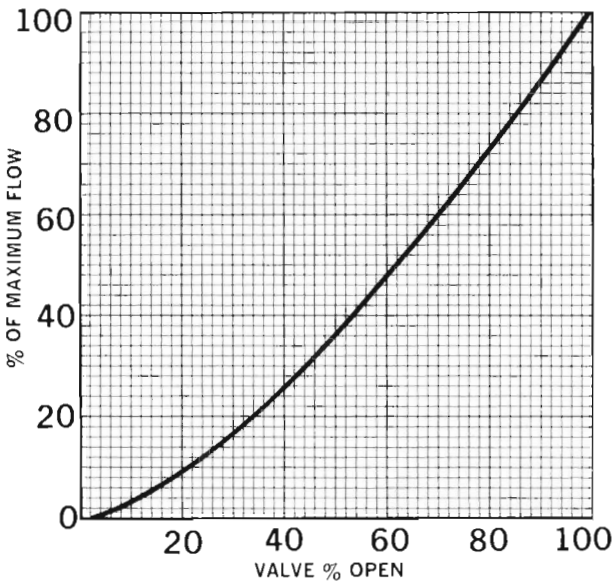


Figure 7. Throttling characteristics.

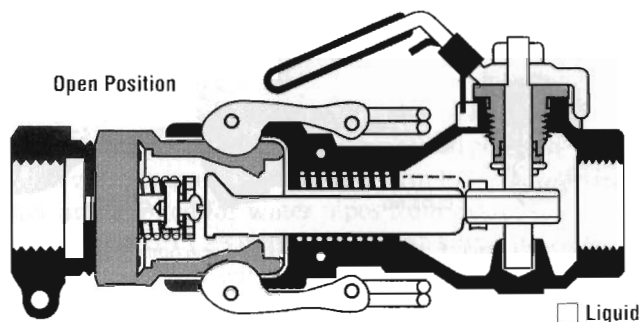


Figure 8. Dry disconnect valve: cutaway view.



Figure 9. Dry disconnect valve.

Polymer valves

Polymer valves are used to drain feed, changeover, sample, inject, distribute and control polyester (including PET), nylon, PVC, PP, PU, HDPE, LDPE and related polymers.

The different types of valves used in the polymer processes include:

- Feeding valves—disc valves are better suited than ram valves for feeding low viscosity feedstocks into reactors (Figure 10). Typical feeding valves are used to regulate the flow of polyester into esterification and polymerization reactors. For vacuum service, bellows are generally used. These should be of the external type due to the possibility of failure in a crystallising environment.
- Discharge valves—ram and disc-type bottom outlet valves are used to drain reactors or control access to the transfer lines between reactors and crystallisers (Figure 11).

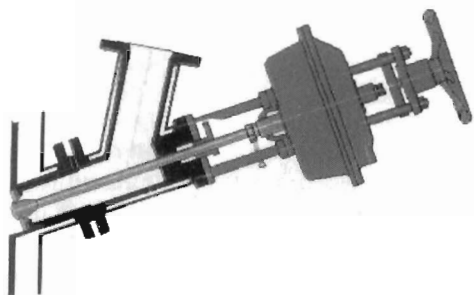


Figure 10. Feeding valve.

- Injection and stripping (deodourising) valve—in many processes, unwanted impurities or remnants, such as unpolymerized monomers in PVC and polyurethanes or solvents in paints and coating suspensions, are stripped from the batch at the end of the process by the injection of saturated steam. The ram tip of the valve is adapted to each vessel as well as the required flow conditions to optimise the spray pattern and prevent product reflux during the operating cycle and in the closed position.

Rather than emptying the reactor when an exothermic reaction goes out of control, the injection and stripping valve (Figure 12) injects a stopper. Another injection valve injects protective colloids. This type of valve is also used to inject steam to heat a reactor.

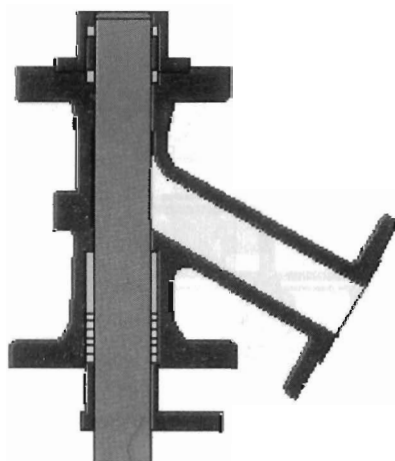


Figure 11. Discharge valve.

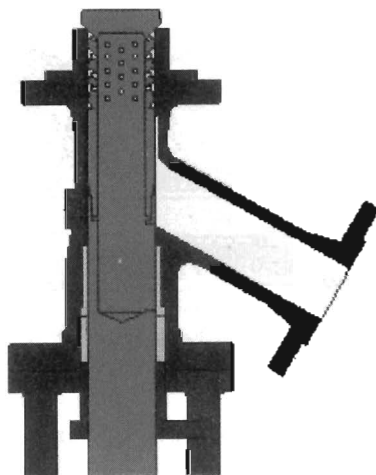


Figure 12. Injection and stripping (desodorising) valves.

- Spray rinse valves—typically used to rinse polymer reactors, especially in the production of PVC where the dome and bottom pad are the parts most in need of cleaning. Valves can be used with water or steam at up to 40 bar.
- Sampling valves—for drain and sampling and flush or purge.
- Polymer additives injection valve—these valves allow small amounts of additives and catalysts such as titanium dioxide to be injected into the line and distributed evenly through the polymer (Figure 13). Different types are available.

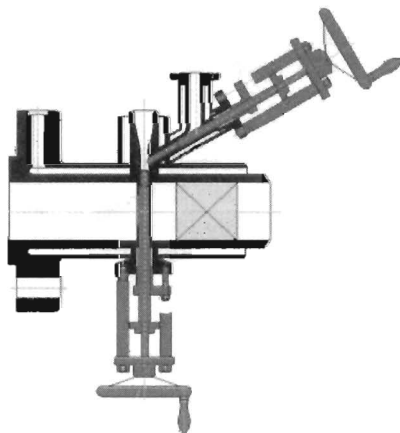


Figure 13. Polymer additives injection valve.

- Multiport diverter valves—these eliminate multiple in-line valves with associated distributor ‘T’ pieces and piping. Main lines can be divided up to six lines with the valves having up to seven inlets and outlets.
- Filter switching valves—usually two valves are involved for switching duplex filters in high-viscosity polymer plants. The inlet valve diverts the flow to one of the duplex filters and the outlet valve guides the flow back to the main line. Piston diverter valves and rotating disc valves are used for this process. Both types of valves can also incorporate particular features and modes of operation.
- In-line valves—these are used to control and shut off the flow of polymer through the piping system. Ball valves and globe valves are not suitable for this duty. Y-globe valves are used in both manual on/off and automatic control versions with polyester. As an alternative, gate valves are used in low pressure and vacuum applications.
- Dye head valves—these valves combine a bottom outlet piston or an end of line Y-globe valve with a stranding dye head, the one used in polyester and nylon polymerization plants.

SECTION 3

Pressure Valves and Services

Check Valves

Safety and Relief Valves

Self-Acting Reducing Valves

Air Relief Valves

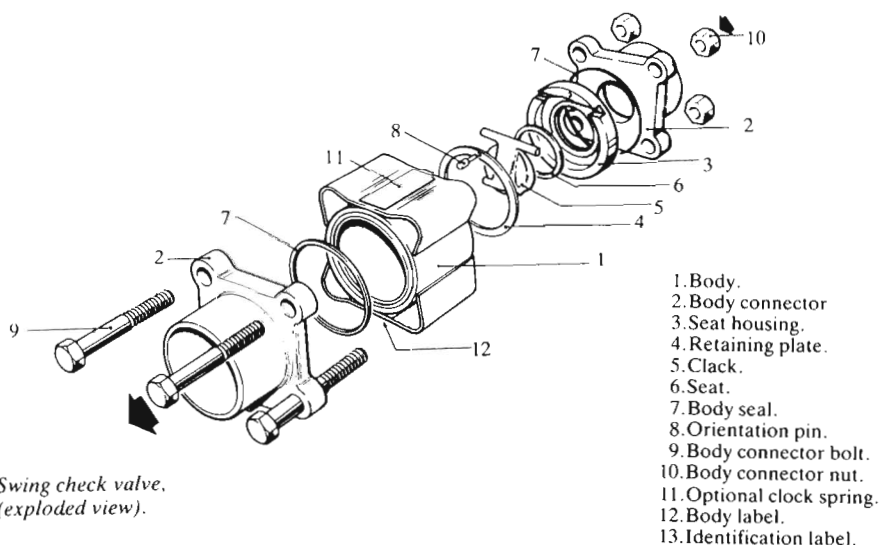
Foot Valves

Check Valves

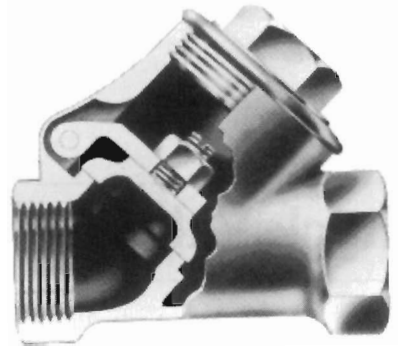
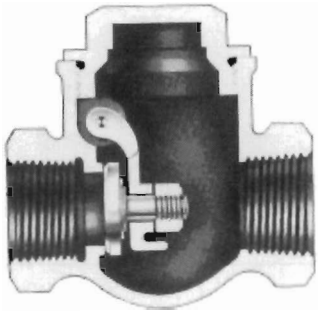
In general terms, check valves are intended to prevent reverse flow in a line e.g. after a pump has stopped and to prevent water hammer. They are also known as non-return valves, reflux valves, flap valves, retention valves and foot valves in different services. The basic principle of the valve is to only allow flow in one direction only and with non-return valves, the check valve is self actuating when flow is reversed. Discs, wafers or membrane diaphragms are used in this type of valve.

There are numerous types of closing systems in check valves but basically the check valve can be categorised as follows:

- (i) Swing- or plate-type valves (swing/plate check valves)—Figure 1, where the check mechanism is a hinged plate or flap, or disc—see chapter on Flap Valves, Section 2. The butterfly check valve is a variant on this principle—see chapter on Butterfly Valves, Section 2.



*Swing check valve.
(exploded view).*



Swing check valves.

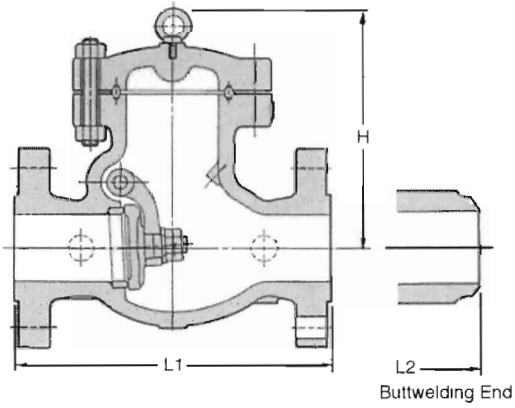


Figure 1. Bolted cover swing check valve.

- (ii) Tilting disc check valves, similar to swing-type check valves but with a profiled disc.
- (iii) Guided or lift-type valves where the check mechanism incorporates an element which lifts along an axis in line with the axis of the body seat. These may be further sub-divided into:
 - (a) disc check valves.
 - (b) piston check valves.
 - (c) ball check valves.
- (iv) Foot valves: specifically check valves fitted to the bottom of a suction pipe.
- (v) Spring-loaded check valves.
- (vi) Wafer check valves: includes swing-type, sprung disc twin plate.
- (vii) Check and surge-suppressor valves: including multi-door check valves for larger pipelines, and electrically- and pneumatically-operated surge-suppressor valves.
- (viii) Hydraulic and pneumatic check valves.

Tilting disc check valves

The basis of the tilting disc check valve is a 'lifting' section disc, pivoted in front of its centre of pressure and counterweighted and/or spring-loaded to assume a normal closed position. With flow in one direction the disc lifts and 'floats' in the stream, offering minimum resistance to flow. The balance of the disc is such that as flow decreases the disc will pivot towards its closed position, reaching this before flow has actually ceased, sealing before reverse flow commences. With reverse-flow, reverse-flow pressure and the counterweight system hold the disc closed (Figure 2). Operation is smooth and silent under all conditions.

Valves of this type normally have resilient sealing rings mounted on a metal face. Metal seals may be used for high-temperature applications.

Guided or lift-type disc valves

Lift-type disc valves are similar in configuration to globe valves except that the disc or plug is automatically operated, i.e. is capable of floating in its seat. The

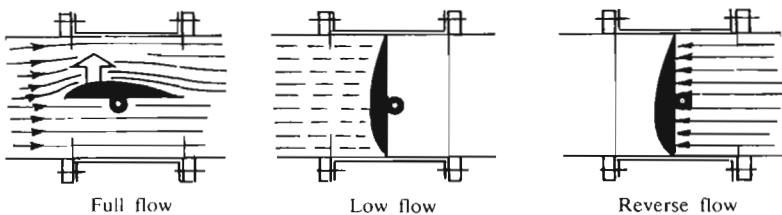


Figure 2. Tilting disc check valve.

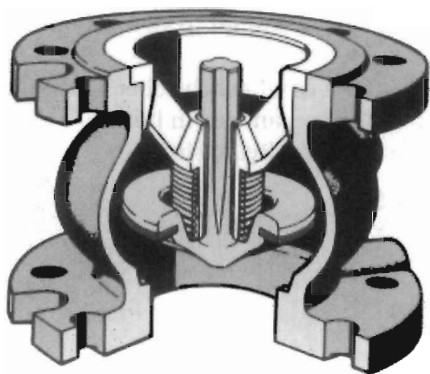


Figure 3. Standard guided non-return check valve.

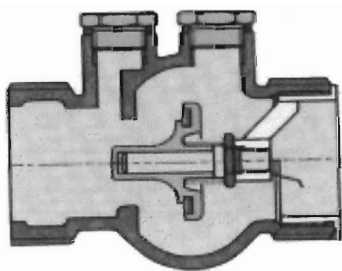
disc or plug is lifted by flow in one direction, permitting through flow. With reverse flow the disc or plug is held on its seat by reverse-flow pressure, giving shut-off. A typical standard check valve is shown in Figure 3.

Valves of this type are further categorised by geometric configuration, i.e. horizontal, angle (oblique) and vertical.

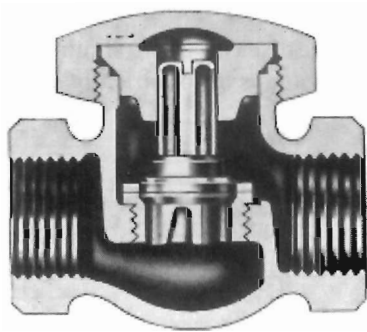
Piston check valves

The piston-type lift check valve incorporates a dashpot applied to the check mechanism (Figure 4), otherwise it is basically similar to a lift-type disc valve. The advantage of the dashpot is that it provides a damping effect during operation.

Lift-type piston check valves are commonly used in conjunction with globe and angle valves on piping systems subject to surge pressures or frequent changes in flow direction.



Check valve for high-duty pressure water systems.



Horizontal lift check valve.

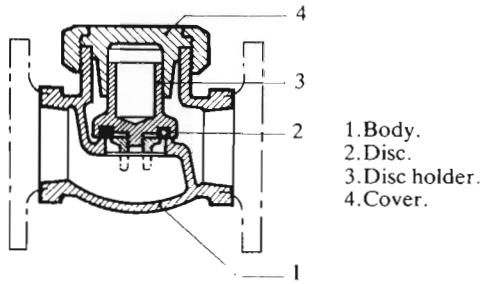


Figure 4. Piston check valve.

Ball check valves

The check element in a lift-type ball valve is a spherical ball, suitably restrained but capable of floating off and onto a seat. With forward flow the ball is forced away from the seat, opening the valve. With reverse flow the ball is forced onto the seat to produce a seal and shut-off. A particular advantage of ball check valves is that they can prove more suitable for use with viscous fluids than other types (Figure 5).

Ball check valves may be of all-metal construction, metal ball with resilient seat, mixed construction (metal or plastic ball), or all-plastic construction.

Foot valves

Foot valves, which often include a strainer, are fitted to the end of a suction pipe and prevent a pump emptying when it stops and therefore not needing priming when restarting. They should have a minimum resistance to flow, with the

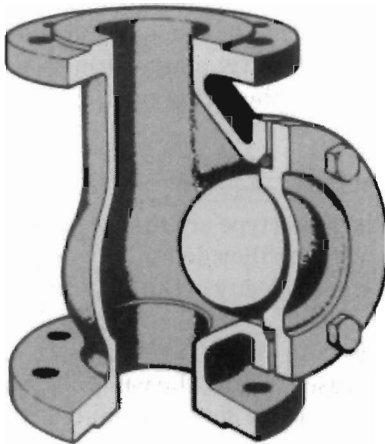


Figure 5. Ball check valve. The closing system is a ball lifted up by the fluid and guided to a lateral housing.

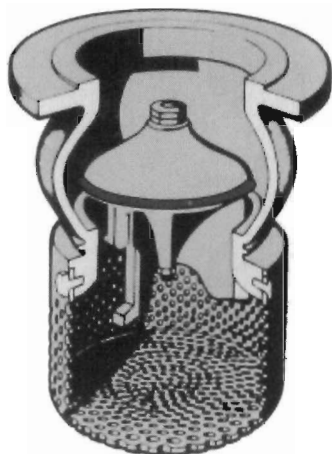


Figure 6. Foot valve for pumping installations with substantial flow.

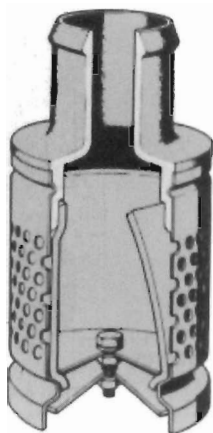


Figure 7. Membrane foot valve for irrigation and drainage pumps.

actual valve element or flap as light as possible if the risk of cavitation is to be avoided. The valves may be of the single flap (Figures 6 and 7) or multiple flat-type, membrane, guided or ball-operating systems. See chapter on Foot Valves.

Spring-loaded check valves

Lift-type check valves may be spring-loaded for more positive shut-off action, particularly as regards more rapid-response cessation of flow, i.e. they can be adjusted to close before flow has fully ceased rather than having to rely on reverse-flow pressure. They can be of disc, plug or ball-type and can work in any position, i.e. horizontal, inclined, upward or downward flow (Figure 8).

Spring-loaded check valves can be made in the widest variety of materials with stainless steel or high-duty alloy springs as necessary. Opening characteristics are governed by the spring rate.

In-line spring assisted valves

The advantages of valves of this type are that they can be installed in the line in any orientation and typically they do not rely on gravity or reverse flow to close. Instead, as the forward velocity of the fluid slows, the spring assist starts to close the disc.

Due to the spring assist and short travel distance of the disc, by the time forward velocity has decreased to zero, the valve disk has reached the seat and the valve is closed. With reverse flow eliminated, the forces necessary to produce water hammer on both upstream and downstream sides of the valve are substantially reduced.

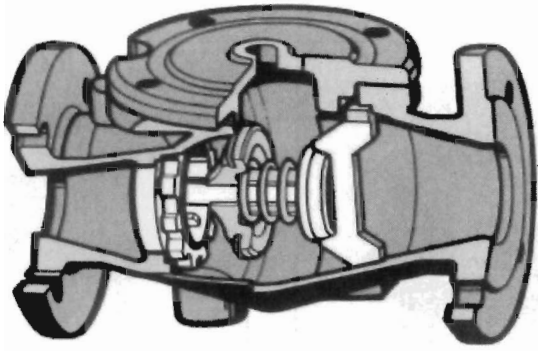


Figure 8. Spring-loaded check valve for protecting drinking water networks.

In-line check valves of this type are probably among the most popular types and are used in many industries including chemical, food and beverage, mining, oil and gas, pulp and paper, building services and general industry duties.

A basic in-line check valve is shown in Figure 9.

A list of typical applications for spring-assisted in-line check valves is shown in Table 1.

Water hammer

This is the generation and effect of high-pressure shock waves (transients) in relatively incompressible fluids. Water hammer is caused by the shock waves that are generated when a liquid is stopped abruptly in a pipe by an object such as a valve disc. Symptoms include noise, vibration and hammering pipe

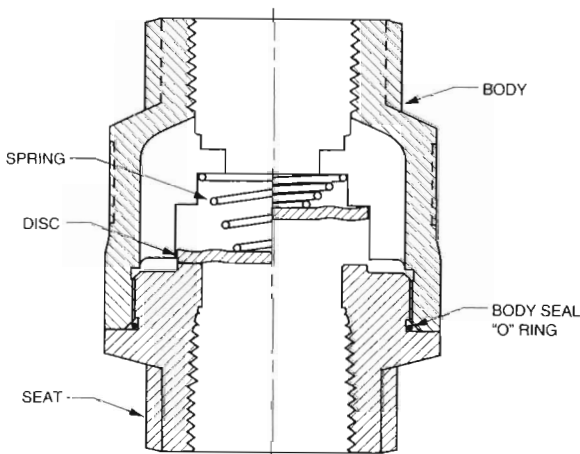


Figure 9. Spring-assisted in-line check valve.



In-line check valves.

sounds which can result in flange breakage, equipment damage, ruptured piping and damage to pipe supports. Whenever incompressible fluids exist in a piping system, the potential exists for water hammer. The risks of water hammer developing are particularly high when the velocity of the fluid is high, there is a large mass of fluid moving and/or when there are large elevation changes within the piping systems. The check valve shown in Figure 10 is specially designed for use on the discharge side of reciprocating air or gas compressors. It includes a pulse damping chamber to maintain the disc in the open position during the momentary reductions in flow associated with each cycle of a reciprocating compressor and to protect against premature seat wear.

Restrictor check valves are generally used for applications that require higher cracking pressures to open the check valve. They should not be considered a substitute for a pressure-relief valve.

A general check valve trouble shooting guide is given in Table 2.

The operation of in-line check valves is not normally affected by their proximity to elbows, 'Ts' control valves, etc. It is not good practice to install

Table 1. In-line check valves

Applications	
<i>Building maintenance</i>	<i>Petroleum production and refining</i>
Compressor discharge	Pump discharge
Condensate lines	Steam lines
Pump discharge	Vacuum lines and breakers
Steam lines	Water treatment
Water lines	
<i>Chemical processing</i>	<i>Power generation</i>
Boiler feed and discharge	Boiler feed and discharge
Compressor discharge	Compressor discharge
Condensate lines	Cooling towers
Cooling towers	Evaporators
Cryogenics	Fly ash system
Evaporators	Pump discharge
Metering pumps	Steam lines
Mineral dewatering	Vacuum system
Nitrogen purge	Water lines
Process lines	
Pump discharge	<i>Primary metals</i>
Steam lines	Chemical lines
Vacuum lines and breakers	Compressor discharge
Water treatment	Condensate lines
	Evaporators
<i>Food, beverage and drug</i>	Extrusion equipment
Autoclaves	Hydraulic lines
Boiler feed and discharge	Presses — water inlet and outlet
Chemical lines	Pump discharge
Compressor discharge	Steam lines
Condensate lines	Water lines
Cookers	Water treatment
Evaporators	
Metering pumps	<i>Pulp and paper</i>
Pump discharge	Boiler feed and discharge
Refrigeration (hot gas defrost)	Chemical lines
Steam lines	Condensate lines
Vacuum lines and breakers	Generator inlet and discharge
	Metering pumps
<i>Mining</i>	Pump discharge
Boiler feed and discharge	Steam lines
Mine dewatering	(digester and paper machines)
	Water treatment
<i>Petroleum production and refining</i>	
Boiler feed and discharge	<i>Textiles</i>
Compressor discharge	Boiler feed and discharge
Condensate lines	Chemical dye lines
Cooling towers	Compressor discharge
Crude and refined product lines	Condensate lines
Evaporators	Metering pumps
Generator inlet and discharge	Pump discharge
	Steam lines

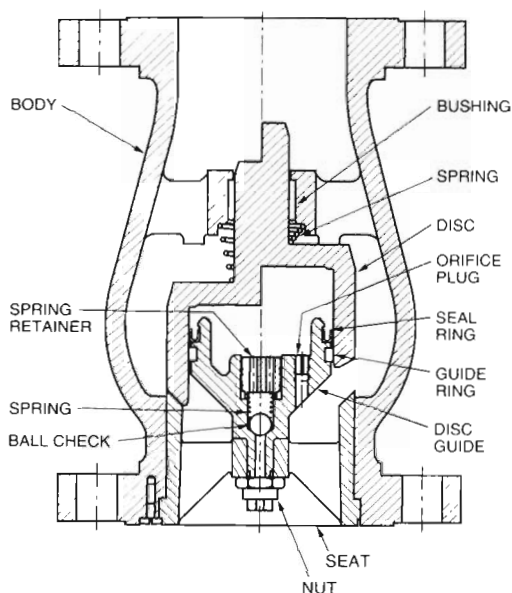


Figure 10. Check valve for reciprocating compressors.

in-line check valves directly to the outlet of such devices as it can result in decreased life due to turbulence caused by the fitting. Some manufacturers recommend that in-line check valves be installed a minimum of five pipe-diameters downstream of any fitting that would cause turbulence. The flow arrow on the body casing, if shown, must be pointed in the direction of the flow.

Wafer check valves

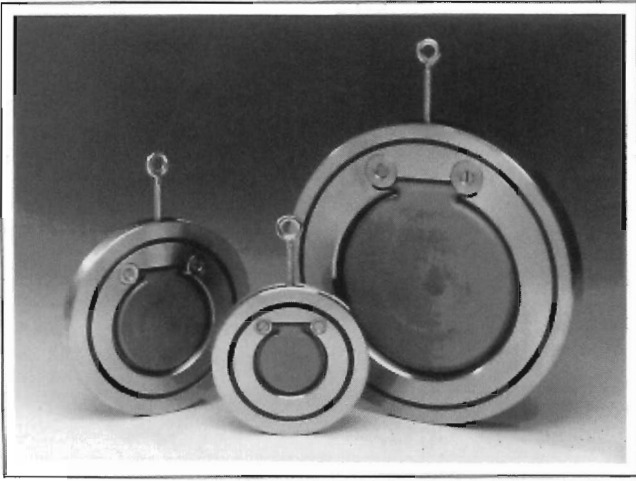
Typically wafer-style-design check valves are used as an effective solution for the prevention of reverse flow in pipes carrying most types of liquids, steam gases, and vapours. They are usually designed to fit between two pipeline flanges.

The valves are opened by the flow pressure of the fluid and closed by a spring when flow ceases and before reverse flow can occur (Figure 11).

Typical applications include:

- Steam boiler flooding protection
- Pipeline fitting protection
- Prevention of reverse flow
- Vacuum breaker

A typical wafer check valve with its pressure-loss diagram is shown in Figure 12.



Water-type check valve.

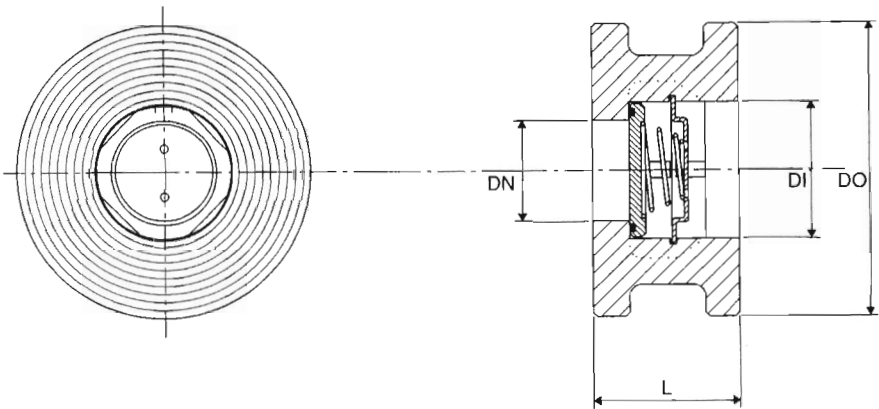
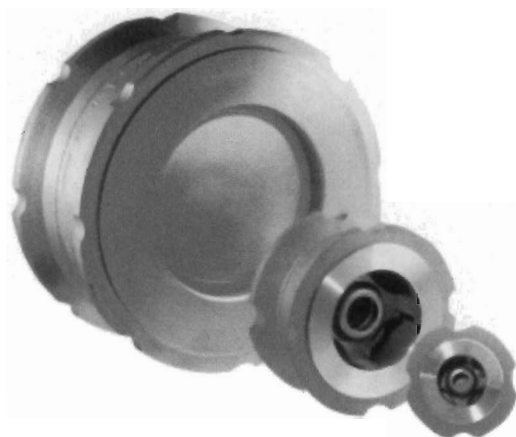


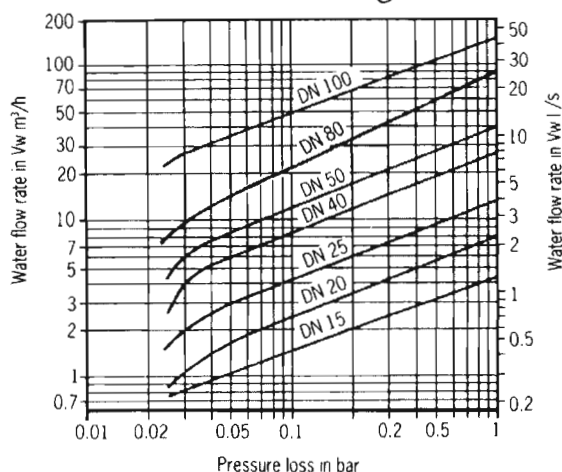
Figure 11. Spring disc wafer check valve.

Hydraulic and pneumatic check valves

Check valves employed in hydraulic and pneumatic applications are more comprehensively covered in the *Hydraulic Handbook* and the *Pneumatic Handbook* both published by Elsevier Science Limited. These valves are generally used where high pressures (up to $10,000 \text{ lb/in}^2$) in standard form where positive leak-tight sealing is required.



Pressure loss diagram



Pressure loss diagram with open valve at 20°C

The values indicated are applicable to spring loaded valves with horizontal flow. With vertical flow, insignificant deviations occur only within the range of partial opening.

The curves given in the chart are valid for water at 20°C. To determine the pressure drop for other fluids the equivalent water volume flowrate must be calculated and used in the graph.

$$V_w = \sqrt{\frac{Q}{1000}} \times V$$

V_w = Equivalent water volume flow in l/s or m³/h

Q = Density of fluid kg/m³

V = Volume of fluid l/s or m³/h

Figure 12. Typical wafer-type check valve and pressure loss diagram.

Table 2. Check valve trouble-shooting guide

Symptom	Cause	Solutions
Water hammer, loud noise, vibration, ruptured piping, equipment damage.	Slow-closing check valve.	In-line spring assisted check valve.
Steam wear (pointed stem), elongated seat guide, bushing wear.	Low flow, pulsating flow, improper sizing.	Custom sizing of the check valve internals. PDC for reciprocating air or gas medium.
Excessive seat leakage (greater than MSS-SP61).	Dirt, trash, foreign substance in the valve.	Clean out the valve. Install strainers if it is a recurring problem. Install a soft seat if bubble tight shut-off is required.
Noise, clicking, tapping.	Low flow, pulsating flow, improper sizing.	Custom sizing of the check valve internals. PDC for reciprocating air or gas medium.
Reverse flow.	Slow-closing check valve.	In-line spring-assisted check valve.
Component breakage, valve failure.	Reciprocating compressor.	PDC for reciprocating air or gas medium.
Missing internals.	Valve not full open, pulsing flow, improper sizing.	Custom sizing of the check valve internals. PDC for reciprocating air or gas medium.

Various types of hydraulic and pneumatic check valves are shown in Figure 13. The distinguishing feature of these valves is their zero leakage achieved by a flexible seal seat (Figure 14).

The flexible seal seat design allows the poppet in the check valve to impact only slightly on the 'O' ring in the closed position. The metal-to-metal contact between the poppet and the end cap serves as a mechanical seat. Under reverse pressure, the 'O' ring flexes only as much as is needed to seal around the nose of the poppet and to expel any foreign particles. As a result the 'O' rings are protected from excessive wear.

Five flow holes drilled into the poppet core are positioned to provide a streamlined flow path through the valve. The combined area of these holes is greater than the area of either the inlet or outlet parts. The flow is directed through the centre of the spring.

Typically, hydraulic and pneumatic check valves incorporate ball-type, poppet, cartridge shuttle and split-flange designs and are used in a wide variety

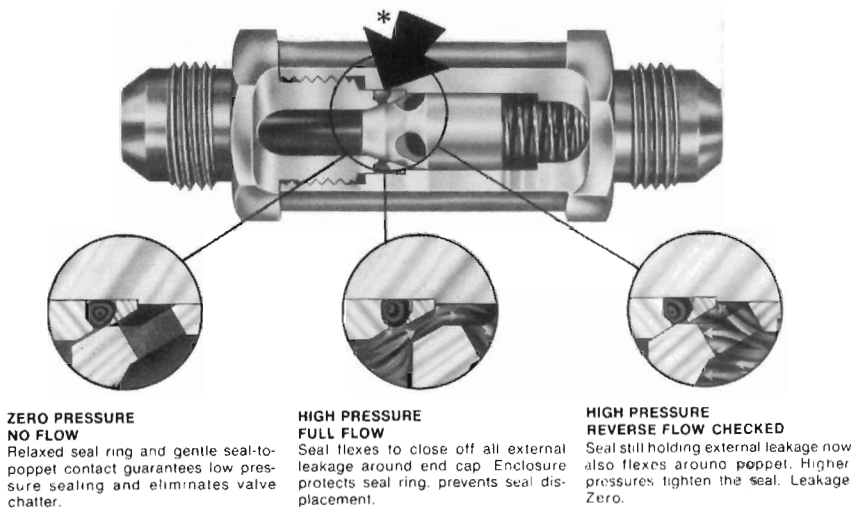


Figure 13. Hydraulic and pneumatic check valve with flexible seal seat design. *Flexible seal seat.



Figure 14. Hydraulic and pneumatic check and relief valves. *Flexible seal seat.

of industries including agriculture, aerospace, road equipment, robotics, industrial machinery, medical equipment instrumentation and controls, chemical processing and handling.

Check valves are commonly used in combination with flow control valves, the type and operating characteristics of which can influence the choice of check valve type.

Suitable combinations are:

Swing check valve—used with ball, plug, gate or diaphragm control valves.

Lift check valve—used with globe or angle valves.

Piston check valve—used with globe or angle valves.

Butterfly check valve—used with ball, plug, butterfly, diaphragm or pinch valves.

Spring-loaded check valves—used with globe or angle valves.

The exception is the foot valve, normally associated with a pump (i.e. there is no other valve positioned between the foot valve and the pump).

See also chapters on Swing check/Flap valves, Non-return valves, Water services.

Safety and Relief Valves

Valves that are vital for the protection of people and plant are termed Safety and Relief Valves. These valves operate automatically when a predetermined pressure level is exceeded by releasing an adjustable spring which holds a valve disc against a valve seat.

There are, however, distinctions between safety valves and relief valves that lead to the following definitions and terminology.

- **Safety Valve**—A valve which automatically discharges gases and vapours so as to prevent a pre-determined safe pressure being exceeded. It is characterised by a rapid full-opening action and used for steam, gases or vapour service (Figure 1).

Safety valves can be further categorised as follows:

- (i) **Low-Lift Safety Valve**—A low-lift valve in which the disc lifts automatically such that the actual discharge area is determined by the position of the disc.
- (ii) **Full-Lift Safety Valve**—A valve in which the disc lifts automatically such that the actual discharge area *is not* determined by the position of the disc.
- (iii) **Pilot-Operated Safety Valve**—A safety valve, the operation of which is initiated and controlled by the fluid discharged from a pilot valve which is itself a direct loaded safety valve.

- **Relief Valve**—A valve which automatically discharges fluid, usually liquid when a pre-determined upstream pressure is exceeded (Figure 2). It may be provided with an enclosed spring housing suitable for closed discharge system application.
- **Pressure Relief Valve**—A safety device designed to protect a pressurised vessel or system during an overpressure event, by relieving excess pressure, and to reclose and prevent the further flow of fluid after normal conditions have been restored. It is characterised by a rapid opening pop action or by opening generally proportional to the increase in pressure over the opening pressure (Figure 3).

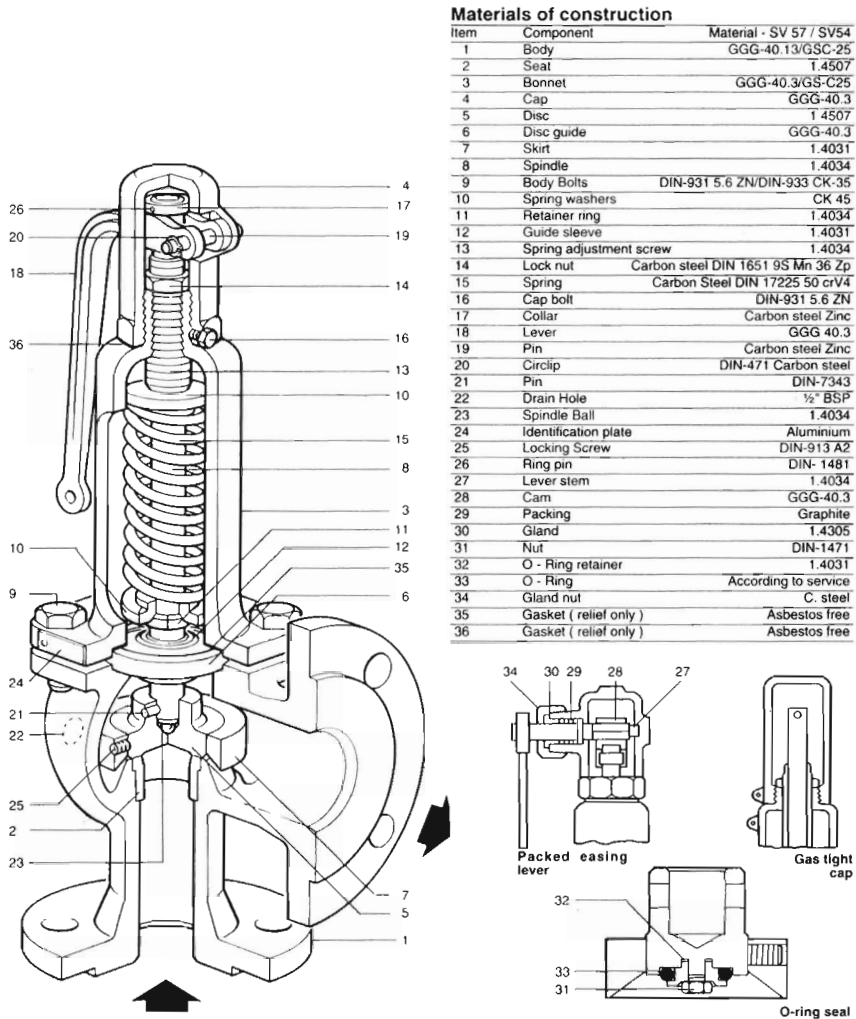


Figure 1. Safety valve suitable for steam, gas and liquid service.

- Pilot-Operated Pressure Relief Valve—A second type of pressure relief valve in which the major relieving device is combined with and is controlled by a self-actuated auxiliary pressure relief valve.
- Safety Relief Valve—A valve which will automatically discharge gases, vapours or liquids so as to prevent a pre-determined safe pressure being exceeded. It is characterised by a rapid full-opening action or by opening in proportion to the increase in pressure over the opening pressure, depending on the application, and may be used either for liquid or compressible fluid (Figure 4).

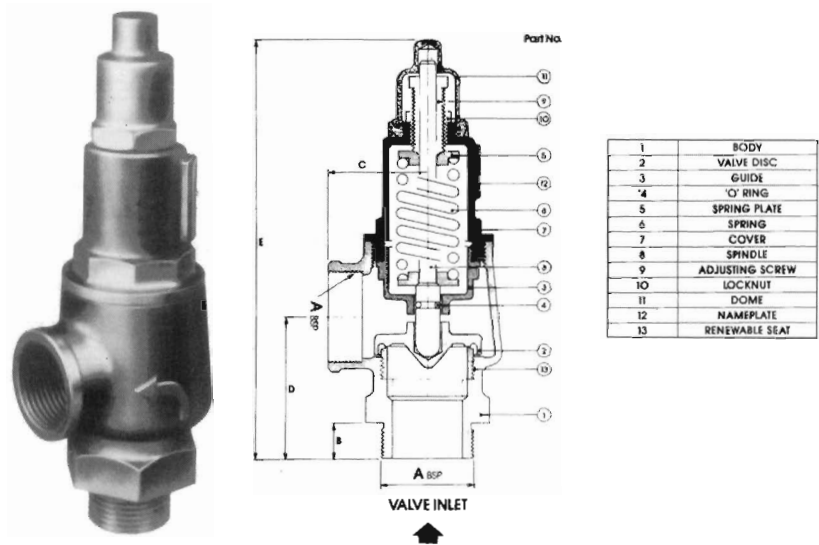


Figure 2. Liquid relief valve.



Figure 3. Pressure relief valve.



Figure 4. Full-lift safety relief valve.

A safety relief valve can be further categorised as:

- (i) A Conventional Safety Relief Valve—A valve which has a spring housing vented to the discharge side of the valve. The operational characteristics (open and closing pressure and relieving capacity) are directly affected by changes of the back pressure on the valve.
- (ii) A Balanced Safety Relief Valve—A valve which incorporates means of minimising the effect of back pressure on the operational characteristics (opening and closing pressure and relieving capacity).

Since all of these types of valves are safety devices, there are many codes and standards throughout the world written to control their design and application. Some of these codes and standards are shown in Table 1. Among the most widely used is the ASME Boiler and Pressure Vessel Code, commonly referred to as the ASME Code.

More specific information may be found by referring to this code, various published standards and by consulting literature published by safety and relief valve manufacturers.

Safety Valves

Typical operating parameters for safety valves are given in Table 2. Safety valve set pressure and temperature limits are governed by a number of factors and may not always coincide with manufacturers' published limits for the applicable materials and flange ratings. Particular limits may be based on spring limitations, specific material selection or other design considerations. With boiler applications, for example, set pressures and total capacity requirements for safety valves are usually established by the design agent or boiler manufacturer.

Safety valves are intended to open and close within a narrow pressure range; therefore, valve installations require careful and accurate design, both as to inlet and discharge piping. The higher the operating pressure and the greater the valve capacity, the more critical becomes the need for proper design of the installation.

Safety valves should always be mounted in a vertical position directly on nozzles having a well rounded approach that provides smooth, unobstructed flow from the vessel or line to the valve.

A safety valve should never be installed on a nozzle having an inside diameter smaller than the inlet connection to the valve, or on excessively long nozzles. The pressure drop occurring in the inlet piping should be calculated at actual flow of the valve. Where safety valves are installed to protect piping systems, as on the low pressure side of a reducing valve or on a turbine by-pass, the pipe or header must be of sufficient size to maintain flow under the safety valve while it is discharging. A typical design of a pop-type safety valve is shown in Figure 5.

Table 1. Codes and standards

Regulatory body	Codes and standards
Allami Energerhkaies Energiabiztonsagtechnikai Felugyelet (AEEF) (State Authority for Energy, Management and Safety) Budapest VIII Koztarsasag ter 7, Hungary	Safety Valves 22/1969/VI.1.2 (mod) 29/1960/VI.7 (orig)
American National Standards Institute 1430 Broadway New York, NY 10018, USA	B16.34 Steel Valves, Flanged and Buttwelded Ends B16.5 Steel Pipe Flanges and Flanged Fittings B31.1 Power Piping B31.3 Chemical Plant and Petroleum Refinery Piping B31.4 Liquid Petroleum Transportation Piping Systems B95.1 Terminology for Pressure Relief Devices ANSI/ASME PTC 25.3 Performance Test Code, Safety and Relief Valves
American Petroleum Institute 2101 L Street Northwest Washington, DC 20037, USA	API RP 510 Pressure Vessel Inspection Code API RP 520 Recommended Practice for the Design and Installation of Pressure Relieving Systems in Refineries: Part I—Design; Part II—Installation API RP 521 Guide for Pressure Relief and Depressuring Systems API Standard 526 Flanged Steel Safety Relief Valves API Standard 527 Commercial Seat Tightness of Safety Relief Valves with Metal to Metal Seats API 2000 Venting Atmospheric and Low Pressure Storage Tanks API Guide for Inspection of Refinery Equipment Chapter XVI—Pressure Relieving Devices
The American Society of Mechanical Engineers United Engineering Center 345 East 47th Street New York, NY 10017, USA	Boiler and Pressure Vessel Code Section I—Power Boilers Section II—Materials Section IV—Heating Boilers Section VII—Care of Power Boilers Section VIII—Pressure Vessels Section IX—Welding and Brazing Qualifications

Table 1 (continued)

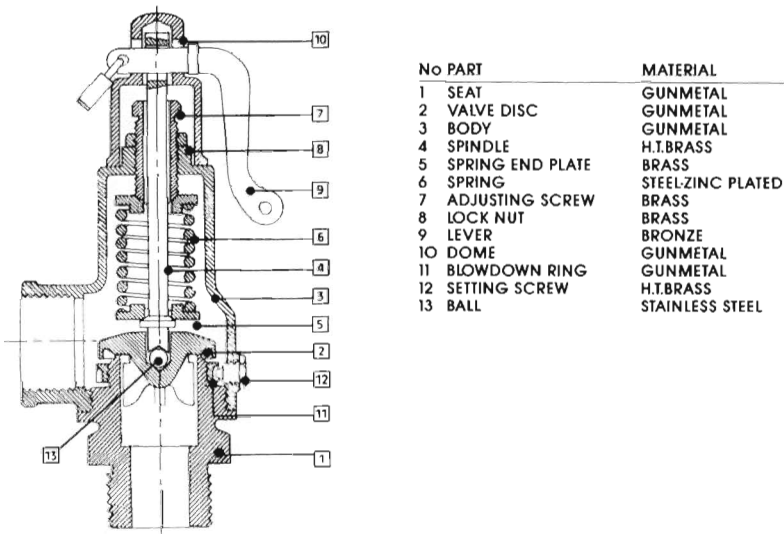
Regulatory body	Codes and standards
Association Francaise de Normalisation Tour Europe Cedex 7 F-92049 Paris La Defence, France	NFE 29-410 to 420
Australian Standards Association No. 1 The Crescent Homebush New South Wales 2140, Australia	AS1271 Safety Valves, Other Valves, Liquid Level Gages and Other Fittings for Boilers and Unfired Pressure Vessels 1990 Edition AS1210 Unfired Pressure Vessels (EAA Unfired Pressure Vessel Code) 1989 Edition AS1200 Pressure Equipment 1994 Edition
British Standards Institute 389 Chiswick High Road London W4 4AL, England	BS6759 Parts 1, 2 and 3 Safety Valves
Canadian Standards Association 178 Rexdale Boulevard Toronto, Ontario M9W 1R3, Canada	CSA Z299.2.85 (R1991)—Quality Assurance Program—Category 1 CSA Z299.3.85 (R1991)—Quality Assurance Program—Category 3 CSA Z299.4.85 (R1991)—Quality Assurance Program—Category 4
Chlorine Institute Inc. 2001 L Street, NW Washington, DC 20036, USA	Pamphlet Type 1-1/2" JQ Pamphlet 41 Type 4" JQ
CC NASTHOL Shenogina Street 123007 Moscow, Russia	GOST R Certification System
Deutsche Institut Fur Normung Burggrafenstrasse 6 D-10787 Berlin, Germany	DIN 50049 Materials Testing Certificates
Comite Europeen de Normalisation (European Committee de Standardisation) rue de Stassart 36 B-1050 Brussels, Belgium	CEN Standards for Safety Valves Pressure Equipment Directive
Heat Exchange Institute, Inc. 1300 Sumner Avenue Cleveland, OH 44115, USA	HEI Standards for Closed Feedwater Heaters
International Organisation for Standardisation Case Postale 56 CH-1211 Geneve 20, Switzerland	ISO-900 Quality System ISO-4126 Safety Valves—General Requirements

Table 1 (*continued*)

Regulatory body	Codes and standards
I.S.C.I.R. Central Bucuresti Frumoasa nr. 26, Romania	Romanian Pressure Vessel Standard
Japanese Industrial Standard Committee Japanese Standards Association 1-24, Akasaka 4-chome, Minato-ku Tokyo 107 Japan	JIS B8210 Spring Loaded Safety Valves for Steam Boilers and Pressure Vessels
Manufacturers' Standardization Society of the Valve and Fitting Industry 1815 North Fort Myer Drive Arlington, VA 22209, USA	SP-6 Finishes of Contact Faces of Connecting End Flanges SP-9 MSS Spot Facing Standard SP-55 Quality Standard for Steel Castings
Ministerie Van Sociale Zaken En Werkgelegenheid Directoraat Generaal Van De Arbeid Dienst Voor Het Stoomwezen 2517 KL Gravenhage Eisenhowerlaan 102, The Netherlands	Stoomwezen Specification A1301
National Association of Corrosion Engineers P.O. Box 1499 Houston, TX 77001, USA	NACE MR0175
National Board of Boiler and Pressure Vessel inspectors 1055 Crupper Avenue Columbus, OH 43229, USA	NB-25 National Board Inspectors Code NB-65 National Board Authorization to Repair ASME and National Board Stamped Safety Valves and Relief Valves
National Fire Protection Association Batterymarc-Park Quincy, MA 02269, USA	NFPA 30 Flammable and Combustible Liquids Code
Schweizerischer Verein für Druckbehälterüberwachung (SDVB) Postfach 35 8030 Zurich, Switzerland	Specifications 602—Safety Valves for Boilers and Pressure Vessels
Den Norske Trykktøiløstkomite (TBK) Norsk Verkstedsindustri Standardiseringsentral Oscarsgate 20, Oslo, Norway	TBK General Rules for Pressure Vessels
Verband der Technischen Überwachungs-Vereine e. V. (TUV) Kurfürstenstraße 56 4300 Essen 1, Germany	TRD 421 AD-Merkblatt A2

Table 2. Safety valve operating parameters

Parameter	Definition
Set pressure (also known as crack pressure)	(i) <i>Liquid services</i> : inlet pressure at which valve starts to discharge under service conditions. (ii) <i>Gas or steam services</i> : inlet pressure at which the valve pops under service conditions.
Differential set pressure	Difference between set pressure and back pressure (where present).
Overpressure	Pressure increase over the set pressure of the valve or relief device.
Blowdown	Difference between the set pressure and resetting pressure expressed either as a specific pressure or percentage of the set pressure.
Back pressure	Any pressure on the discharge side of a pressure-relief valve.
Accumulation	Pressure increase over maximum allowable working pressure of a vessel or system during discharge through the pressure relief valve.
Operating pressure	Normal allowable working pressure of the system or vessel.

*Figure 5. Typical pop-type safety valve.*

The discharge piping from safety valves should be equal in size to, or larger than, the nominal valve outlet and should be as simple and direct as possible. Good practices must be observed with discharge manifold lines. All discharge piping in the discharge system must be vented to a safe disposal area to prevent personnel injury when the valve discharges.

The valve shown in Figure 6a is typically used for steam generators and steam systems. It is a high capacity reaction-type valve designed specifically for saturated steam service on boiler drums having design pressures above 103 bar (1500 lb/in²).

A typical valve operating cycle (Figure 6b) is as follows:

As pressure in, say, a boiler increases to the safety valve set point the valve will pop open. After the valve opens steam passes through a series of annular flow passages (A) and (B) which control the pressure developed in chambers (C) and (D). The excess steam is exhausted through guide ring openings (E) to the valve body bowl (F).

As pressure in the boiler decays, the dynamic forces on the lower face of the disc holder assembly are reduced and the safety valve disc begins to close. Assisted by pressure in chambers (C) and (D), the valve at this point closes sharply and tightly.

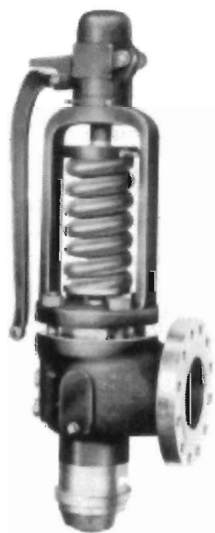


Figure 6A. Safety valve for saturated steam applications.

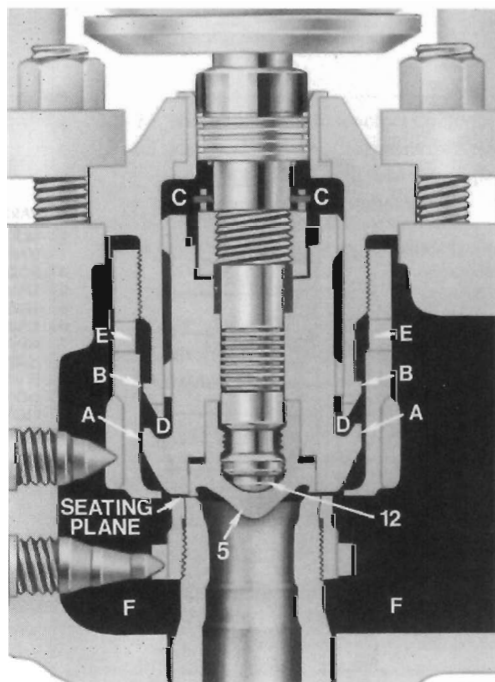


Figure 6B. Safety valve operating principle.

Figure 7 shows a safety valve frequently used in process applications. The valve has a closed bonnet that contains the process fluid within the safety valve preventing any release to atmosphere.

In Figure 8, the safety valve incorporates a balanced bellows to provide satisfactory safety valve performance when the developed back pressure becomes excessive. Balanced bellows ensure that safety valve characteristics such as lift and relieving capacity, opening and closing pressure and stability are not unduly influenced by static pressure in the discharge manifold.

Balanced safety valves must be installed when the percentage build-up back pressure in the exhaust system is allowed to exceed the percentage overpressure applicable to the safety valve. Valves that vent to the atmosphere, either directly or through short vent stacks, are not subjected to elevated back pressure conditions. Valves installed in a closed system, i.e. on corrosive, toxic or valuable recoverable fluids, or when a long vent pipe is used, may develop high back pressure. Back pressure which may occur in the downstream system while the safety valve is closed is called superimposed back pressure. Back pressure which may occur after the valve is open and flowing is called dynamic or built-up back pressure.

Figure 9 shows a double-spring high-lift safety valve that combines a top guided design to provide an unobstructed seat bore with a floating disc.

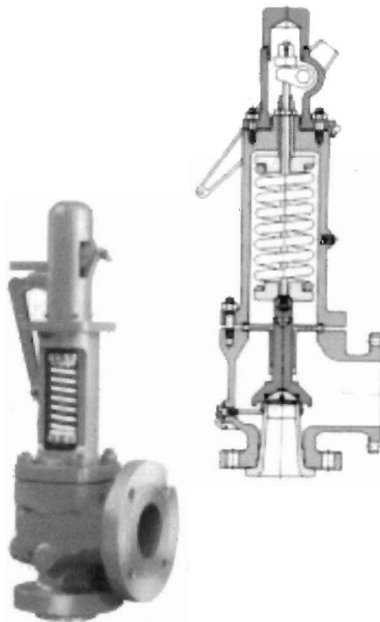


Figure 7. Basic safety valve for process applications (closed bonnet type).

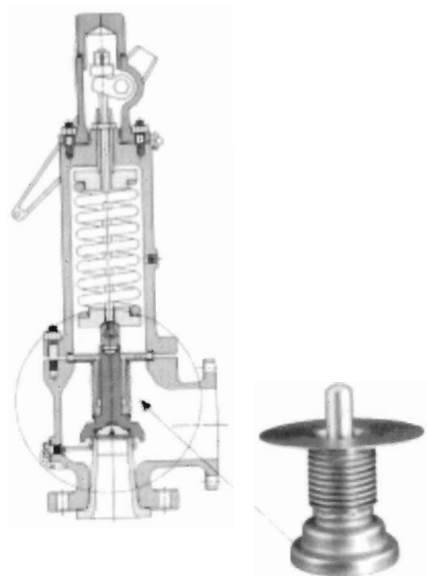


Figure 8. Safety valve for process applications (balanced bellows type).

Relief Valve

A basic difference between the design of spring-loaded safety valves and relief valves is that, in safety valves, the poppet or disc overhangs the seat to promote faster lift whereas, in a relief valve, the area exposed to overpressure is the same whether the valve is open or closed. As a consequence, a safety valve pops open while a relief valve lifts gradually with increasing pressure until it reaches its fully open position.

The relief valve shown in Figure 10 is a standard type suitable for relieving excess pressures of water oil, air, gases or steam where high discharge rates are not required. Duties include the protection of pipelines against overpressure and protection against thermal expansion. It is filtered in the upright position. A spring-loaded side-discharge version is shown in Figure 11.

Other spring relief valves have cartridge-type assemblies for easy cleaning. They are usually suitable for use with positive displacement pumps of the rotary or reciprocating type. They can also be used as combined relief and by-pass valves.

The relief valve (Figure 12) is manufactured from plastic —PVC, PVDF and CPVC with solid Teflon[®]* shaft, intended as a chemical-resistant relief valve for corrosive and pure liquids. The relieving pressures can be adjusted by screwing the adjusting bolt up or down to decrease or increase the pressure setting.

This type of valve is *not* a pop safety-type valve.

*Dupont registered trade mark.

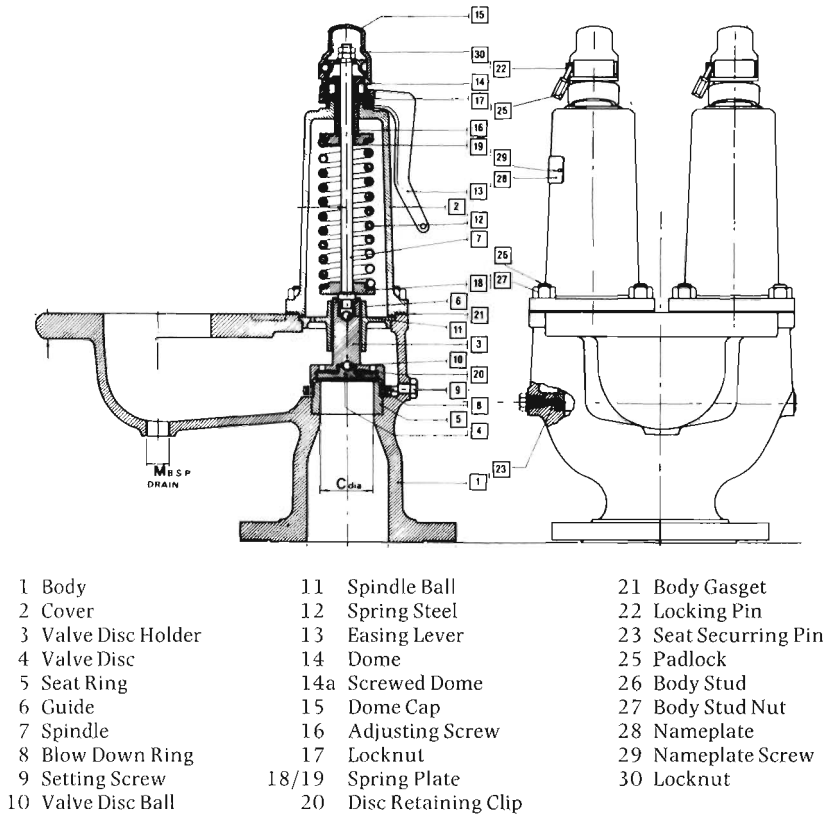


Figure 9. Double-spring high-lift safety valve.

Emergency relief valves of the type shown in Figure 13 are designed to meet the stringent conditions of container, rail, road and static tanks for emergency venting under total fire engulfment conditions. Usually manufactured from 316 stainless steel, these types of valves can incorporate a manually operated vacuum vent button.

The type of relief valves shown in Figure 14 is ideally suited for air, acetylene, ammonia, freon 12 and 22, hydrogen, carbon dioxide, oxygen, aromatic fuels, synthetic oils, tetrachloride and toluene, at operating pressures up to 2400 lb/in². It can be mounted in horizontal and vertical positions.

In closed operation, the spring load is carried by a metal-to-metal seat. An O-ring provides a tight seal and the sealing efficiency increases as the pressure increases up to cracking pressure. At cracking pressure the ports in the poppet open fully and eliminate the rapid increase in pressure. Flow is throttled between the poppet shoulder and seat and a regularly increasing flow area is



Figure 10. Standard spring-operated relief valve.

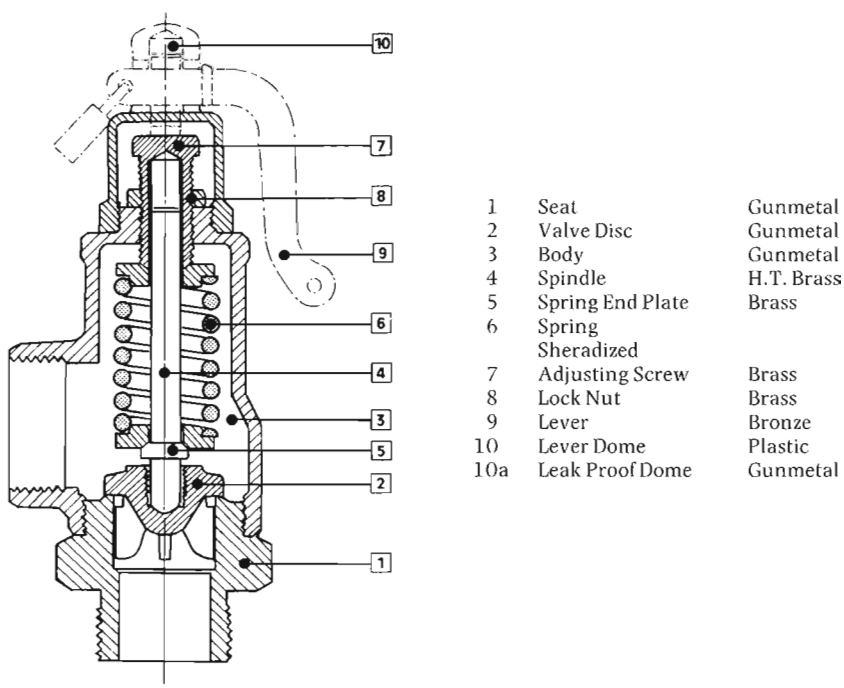


Figure 11. Spring-loaded side-discharge relief valve.

provided with increasing flow rates. When fully open, the inline construction and full flow ports permit maximum flow with minimum increase in system pressure (see Figure 15).

High pressure variants operate at pressures up to 10,500 lb/in².

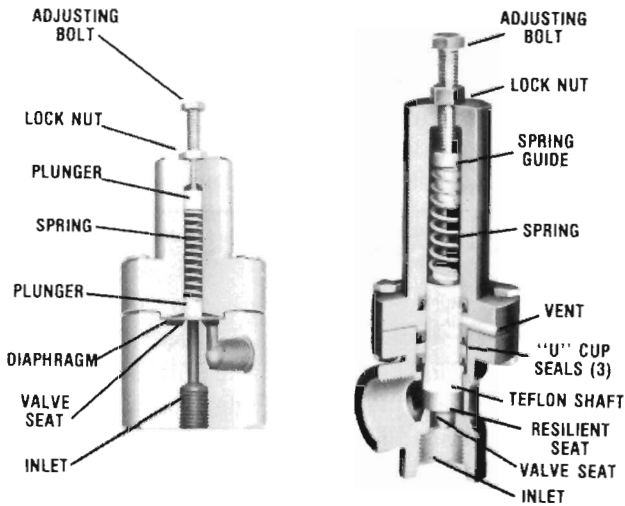


Figure 12. Chemical-resistant relief valves.



Figure 13. Emerging relief valves.

Pressure Relief Valves

The basic spring-loaded pressure relief valve (Figure 16) has been developed to provide overpressure protection.

- Overpressure may be defined as a pressure increase over the set pressure of a pressure relief valve, usually expressed as a percentage of set pressure.
- Set pressure is the pressure measured at the valve inlet at which a pressure relief valve should commence to lift under service conditions.
- Popping pressure is the value of increasing static pressure at which the disc moves in the opening direction at a faster rate as compared with corresponding movement at higher or lower pressures.

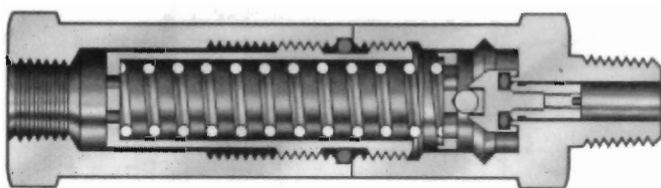
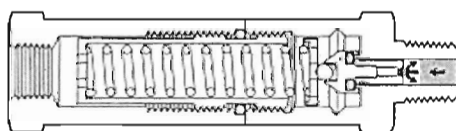
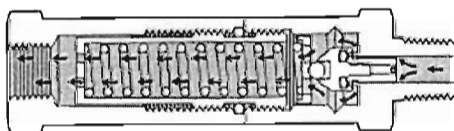


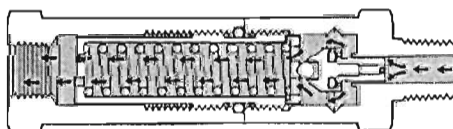
Figure 14. High-pressure relief valve.



CLOSED



CRACKING



OPEN

Figure 15. High-pressure relief valve method of operation.

The valve shown in Figure 16 consists of a valve inlet or nozzle mounted on the pressurised system, a disc held against the nozzle to prevent flow under normal system operating conditions, a spring to hold the disc closed, and a body or bonnet to contain the operating elements. The spring load is adjustable to vary the pressure at which the valve will open.

The sole source of power for the pressure relief valve is the process fluid.

The pressure relief valve must open at a pre-determined set pressure and close when the system pressure has returned to a single safe level. Pressure relief valves must be designed with materials compatible with many process fluids from simple air and water to the most corrosive media. This type of valve is required to remain on systems for long periods of time and must have the ability to maintain tight shut-off. Most manufacturers recommend that system operating pressures not exceed 95% of set pressure to achieve and maintain proper seat tightness integrity.

Examples of spring-loaded pressure relief valves are given in Figure 17.

A rupture disk device (Figure 18) is a non-reclosing pressure relief device actuated by inlet static pressure and designed to function by the bursting of a pressure containing disk. A rupture disk is the pressure containing and pressure sensitive element of a rupture disk device. These products provide full opening with instantaneous pressure relief in the event of system upset. Application of rupture disk devices to liquid service should be carefully evaluated, especially if used in combination with a safety or safety relief valve.

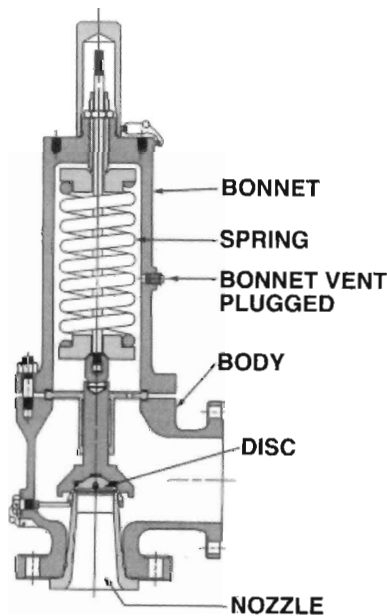


Figure 16. Standard spring-loaded pressure relief valve.

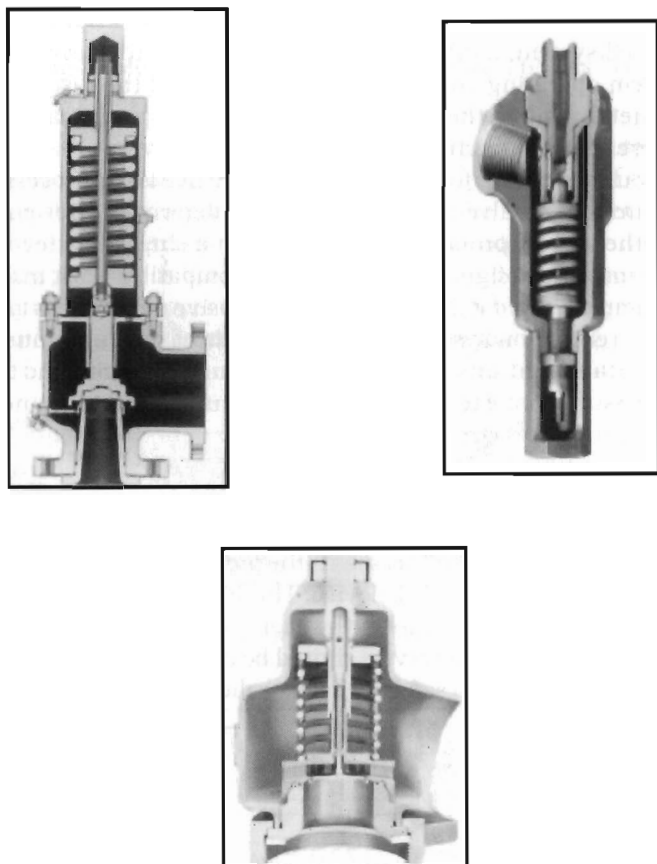
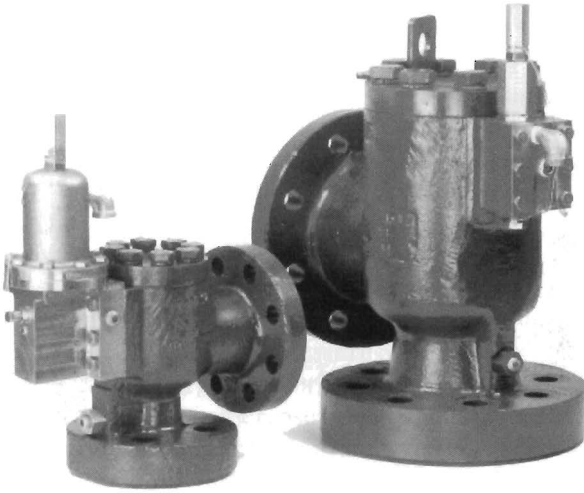


Figure 17. Pressure relief valves: top left—standard valve; top right—screwed valve with single trim; lower—sanitary valve for foodstuffs and pharmaceuticals.

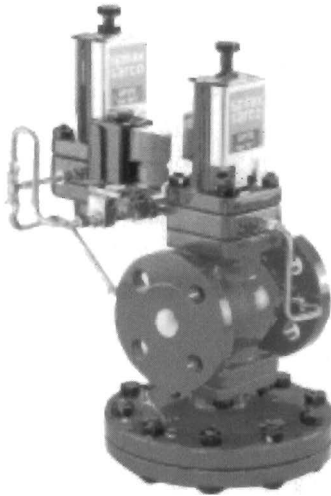
Pilot-Operated Pressure Relief Valve

This type of valve consists of a main valve with a piston- or diaphragm-operated disc and a pilot. Under normal conditions the pilot allows system pressure into the piston chamber. Since the piston area is greater than the disc seat seal area, the disc is held closed. When the set pressure is reached, the pilot actuates to shut off system fluid to the piston chamber and simultaneously vents the piston chamber. This causes the disc to open (Figure 19).

Another version of a pilot-operated pressure reducing valve is shown in Figure 20. Other constructions have integral porting, eliminating the need for tubing to activate the valve and relieve the system pressure, as all pressurisation is performed through porting machined into the main valve and the mating pilot valves.



High-pressure pilot-operated safety relief valve.



Two pilots combined on the main body of a pressure-reducing valve.

Pilot-operated relief valves have several advantages. As the system pressure increases, the force holding the disc in the closed position increases. This allows the system operating pressure to be increased to values within 5% of set pressure without danger of increased leakage in the main valve. Valves can be set fully open at the set pressure and closed with a very short blowdown. A reducing valve will modulate from its maximum capacity down to zero load when it will shut. However, if the valve is to work under low load conditions for much of its life, there may be a good case for fitting two smaller valves in parallel.

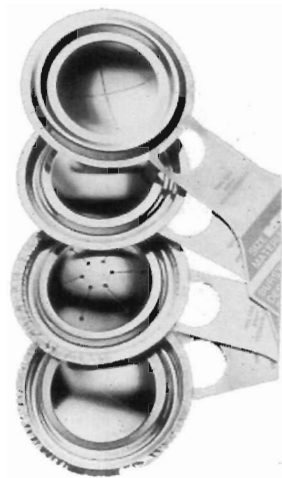


Figure 18. Rupture disk device.

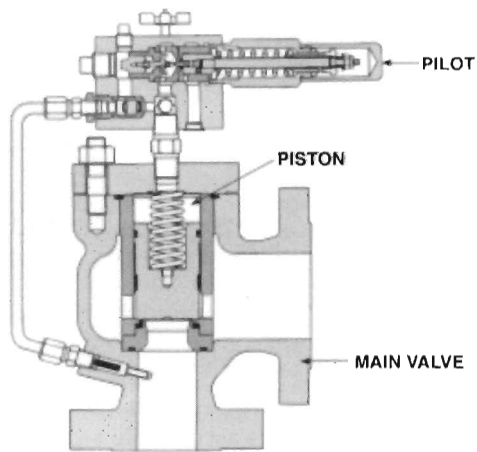


Figure 19. Snap-acting pilot-operated pressure relief valve.

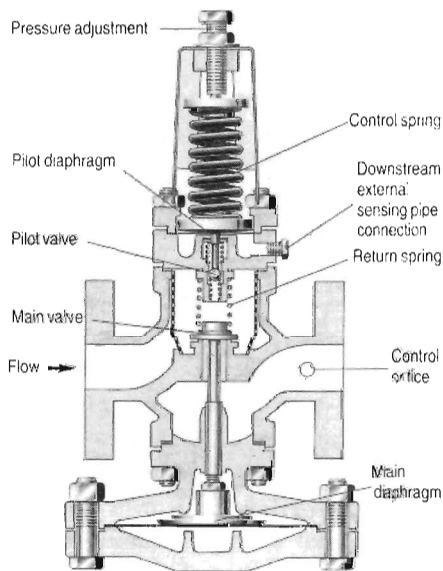


Figure 20. Standard pilot-operated pressure-reducing valve for steam, air and industrial gases.

Modulating pilot valve designs limit fluid loss and system shock. However, this type of valve is generally only recommended for clean service and is found in a broad range of applications and industries including steam, air and industrial gases, petroleum-refining offshore applications, chemical processing, pulp and paper mills and general manufacturing. There are numerous styles and designs available from many manufacturers.

Safety Relief Valve

The purpose of a safety relief valve is to discharge a given amount of vapour, gas or liquid, whilst preventing the pressure increase exceeding a pre-determined level. The safety relief valve should close with the smallest drop in pressure consistent with tight closure, and it should remain pressure-tight up to the time of the next response to an overpressure situation. A standard safety relief valve is shown in Figure 21.

The valve must be reliable so that the action is always a repeat of the previous action. A safety relief valve should be used on any closed vessel or system in which the pressure can be other than atmospheric and where, under any circumstances, the design pressure of the system can be exceeded.

In most instances the discharge pipework is direct to atmosphere, but when the medium is toxic, inflammable or otherwise objectionable, complex-type discharge pipework systems are used and, frequently, more than one valve discharges into the system, resulting in a variable back pressure at the safety relief valve.

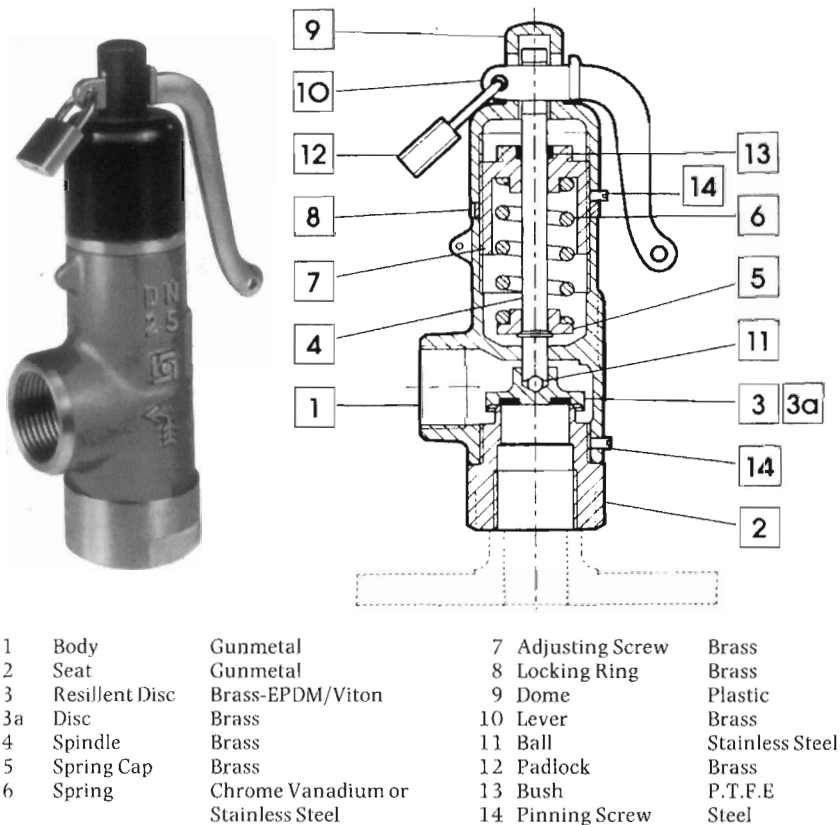


Figure 21. Typical certified safety relief valve.

When such a discharge system is adopted, the safety relief valve must be designed in such a way that the effects of the variable back pressure on the set pressure are minimised. This requires the use of a balanced bellows valve.

The safety relief valve should be as maintenance-free as possible.

Sizing Safety and Relief Valves

Proper sizing and selection of safety and relief valves is critical. The first step in applying overpressure protection to a vessel or system is to determine the type of fluid, set pressure, back pressure, allowable overpressure and required relieving capacity; the next step is to establish inlet temperature, compressibility factor, gas constant or isentropic coefficient, molecular weight, specific weight, specific gravity and viscosity.

Sizing equations are available from manufactures and regulatory bodies, e.g. British Standards BS 6759, American Standards to ASME Code Section VIII and European Standards A.D. Merkblatt A2. All capacities can be calculated in accordance with the internationally accepted sizing equations using the certified coefficients of discharge. Typical sizing equations in accordance with specific standards are, for example:

In accordance with **BS 6759: 1984** Modified for the effects of set pressure below
3.0 bar, back pressure and superheated steam

For steam:

$$A = \frac{E}{0.525 P \cdot K_{dr} \cdot F_p \cdot F_b \cdot F_{SH}} \quad \text{or} \quad E = 0.525 P \cdot A \cdot K_{dr} \cdot F_p \cdot F_b \cdot F_{SH}$$

For compressed air:

$$A = \frac{Q_1}{0.193 P \cdot K_{dr} \cdot F_p \cdot F_b \cdot \sqrt{\frac{288}{T}}} \quad \text{or} \quad Q_1 = 0.193 P \cdot A \cdot K_{dr} \cdot F_p \cdot F_b \cdot \sqrt{\frac{288}{T}}$$

For gases:

$$A = \frac{Q_2}{P \cdot C \cdot K_{dr} \cdot F_p \cdot F_b \cdot \sqrt{\frac{M}{ZT}}} \quad \text{or} \quad Q_2 = P \cdot A \cdot C \cdot K_{dr} \cdot F_p \cdot F_b \cdot \sqrt{\frac{M}{ZT}}$$

For liquids:

$$A = \frac{Q_2}{1.61 K_{dr} \cdot F_v \cdot F_w \cdot \sqrt{\rho \Delta P}} \quad \text{or} \quad Q_2 = 1.61 A \cdot K_{dr} \cdot F_v \cdot F_w \cdot \sqrt{\rho \Delta P}$$

For hot water:

$$A = \frac{Q_3}{0.329 K_{dr} \cdot P \cdot H_{ws}} \quad \text{or} \quad Q_3 = 0.329 A \cdot K_{dr} \cdot P \cdot H_{ws} \cdot \sqrt{\rho \Delta P}$$

Key to equations

A	=	Orifice area	mm ²
E	=	Required capacity of steam	kg/hr
Q ₁	=	Required capacity of compressed air	l/s
Q ₂	=	Required capacity of gas/liquid	kg/hr
Q ₃	=	Required capacity of hotwater	kW
P	=	Absolute inlet pressure (set pressure + overpressure + 1.013)	bar abs
ΔP	=	Relieving pressure – Back pressure (set pressure + overpressure – back pressure)	bar gauge
T	=	Inlet temperature	°K (°C + 273)
ρ	=	Liquid density	kg/m ³
M	=	Molecular weight	kg/kmol
Z	=	Compressibility factor	

$$Z = \frac{10^5 P.V.M.}{R.T.}$$

where

R	=	Universal gas constant—8314 N.m./kmol.k
V	=	Specific volume of gas at STP conditions
C	=	Gas constant

Use the following formula

$$C = 3.9458 \sqrt{K \left(\frac{2}{K+1} \right)^{\frac{K+1}{K-1}}}$$

K	=	Isentropic exponent at the relieving inlet conditions. If the value of K is not available at these conditions the value at 1.0135 bar abs and 15°C should be used
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Capacity correction factors

F _p	=	Capacity correction factor for the effects of low set pressure
F _b	=	Capacity correction factor for the effects of back pressure (balanced bellows valves only)
F _v	=	Capacity correction factor for the effects of viscosity (liquids only)
F _w	=	Capacity correction factor for the effects of back pressure (liquids only; balanced bellows valves only)
F _{sh}	=	Capacity correction factor for the effects of superheat
F _{dr}	=	De-rated coefficient of discharge. Select K _{dr} appropriate to the fluid
F _{ws}	=	Hotwater correction for vented system (0.5 if vented/1 if pressurised)

In accordance with **A.S.M.E. Code Section VIII and API RP 520** Modified for the effects of set pressure below 3.0 bar, back pressure and superheated steam

Note:- When sizing valves in accordance with ASME Code Section VII, the certified capacity may only be calculated at 10% overpressure or 3 psig, whichever is the greater and without the use of correction factors F_p, F_b, F_v, F_w and F_{sh}. Set pressures below 15.0 psig may not be ASME stamped.

For steam:

$$A_1 = \frac{W}{51.5 P_g K_{dr} F_p F_b F_{SH}} \quad \text{or} \quad W = 51.5 P_g A_1 K_{dr} F_p F_b F_{SH}$$

For air/gases:

$$A_1 = \frac{Q \sqrt{G T_g Z}}{1.175 P_g C_g K_{dr} F_p F_b} \quad \text{or} \quad Q = \frac{1.175 P_g C_g A_1 K_{dr} F_p F_b}{\sqrt{G T_g Z}}$$

For liquids:

$$A_1 = \frac{V_L \sqrt{G}}{38 \sqrt{\Delta P_L} K_{dr} F_v F_w} \quad \text{or} \quad V_L = \frac{38 A_1 \sqrt{P_L} K_{dr} F_v F_w}{\sqrt{G}}$$

Key to equations

A_1	= Office area	in ²
Q	= Required capacity of air/gases	SCFM
W	= Required capacity of steam	lb/hr
V_L	= Required capacity of liquid	usgpm
P_g	= Absolute inlet pressure (set pressure + overpressure + 14.7)	psia
ΔP_L	= Relieving pressure – Back pressure (set pressure + overpressure – back pressure)	psig
T_g	= Inlet temperature	°R (°F + 460)
G	= Specific gravity	
Z	= Compressibility factor for the gas or vapour at P or T conditions (if not given use $Z = 1$)	
C_g	= Imperial gas constant. Use the following formula	

$$C_g = \sqrt{K \left(\frac{2}{K+1} \right)^{\frac{K+1}{K-1}}}$$

K	= Isentropic exponent at the relieving inlet conditions. If the value of K is not available at these conditions the value at 14.7 psi abs and 59°F should be used
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Capacity correction factors

F_p	= Capacity correction factor for the effects of low set pressure
F_b	= Capacity correction factor for the effects of back pressure (balanced bellows valves only)
F_v	= Capacity correction factor for the effects of viscosity (liquids only)
F_w	= Capacity correction factor for the effects of back pressure (liquids only; balanced bellows valves only)
F_{sh}	= Capacity correction factor for the effects of superheat
K_{dr}	= De-rated coefficient of discharge. Select K_{dr} appropriate to the fluid

In accordance with **A.D. MERKBLATT A2** Modified for the effects of set pressure below
3.0 bar, back pressure and superheated steam

For steam:

$$A_o = \frac{Q_m \cdot \chi}{K_{dr} \cdot P \cdot F_b \cdot F_p} \quad \text{or} \quad Q_m = \frac{A_o \cdot K_{dr} \cdot P \cdot F_b \cdot F_p}{\chi}$$

For air/gases:

$$A_o = \frac{0.1791 Q_m}{\psi K_{dr} \cdot P \cdot \sqrt{\frac{M}{ZT}} F_p \cdot F_b} \quad \text{or} \quad Q_m = \frac{A_o \cdot \psi K_{dr} \cdot P \cdot \sqrt{\frac{M}{ZT}} F_p \cdot F_b}{0.1791}$$

For liquids:

$$A_o = \frac{0.6211 Q_m}{K_{dr} \cdot \sqrt{\rho \Delta P} F_v \cdot F_w} \quad \text{or} \quad Q_m = \frac{A_o \cdot K_{dr} \cdot \sqrt{\rho \Delta P} F_v \cdot F_w}{0.6211}$$

Key to equations

A_o	= Office area	mm ²
Q_m	= Required capacity of steam, air, gases or liquids	kg/hr
P	= Absolute set pressure (set pressure + 1.013)	bar abs
ΔP	= Relieving pressure – Back pressure (set pressure – back pressure)	bar gauge
T	= Inlet temperature	°K (°C + 273)
ρ	= Liquid density	kg/m ³
M	= Molecular weight	kg/kmol
K	= Isentropic exponent at the relieving inlet conditions. If the value of K is not available at these conditions the value at 1.0135 bar abs and 15°C should be used	
ψ	= A.D. Merkblatt outflow function for air and gases	
χ	= A.D. Merkblatt pressure medium coefficient for steam	
	$= \frac{0.6211 \sqrt{PV}}{\psi}$	

V = Specific volume in m³/kg for supercritical pressure relief

Capacity correction factors

F_p	= Capacity correction factor for the effects of low set pressure
F_b	= Capacity correction factor for the effects of back pressure (balanced bellows valves only)
F_v	= Capacity correction factor for the effects of viscosity (liquids only)
F_w	= Capacity correction factor for the effects of back pressure (liquids only; balanced bellows valves only)
K_{dr}	= De-rated coefficient of discharge. Select K_{dr} appropriate to the fluid

Manufacturers' engineering support information and technical data are essential if the correct valve is to be selected for the job. Some manufacturers supply full details on a computer program. Generally the program is easy to use with many features including: quick and accurate calculations, user-selected units, selection of valve size and style, valve data storage, printed reports, specification sheets and dimensional drawings.

There is no substitute for qualified engineering analysis, and the application of safety and relief valves of any type or make should be assigned only to fully trained personnel and be in strict compliance with rules provided by the governing codes and standards. Selection of safety and relief valves should not be based on arbitrarily assumed conditions or incomplete information. Valve selection and sizing is the responsibility of the system engineer and the user of the equipment to be protected.

Testing Safety Valves

Safety valves should be tested regularly to ensure that they have retained their capability of operating at design lift-off pressure. Two basic methods of testing are:

On-line testing by deliberate overpressure of the system to determine the actual pressure at which the valve lifts off its seat.

Off-line testing by removal of the valve from its line or position and determination of active lift-off by hydraulic test.

It is also possible to apply a hydraulic test for on-line testing using a portable hydraulic test pack.

Self-Acting Reducing Valves

Self-acting reducing valves generally fall into two main categories: (i) direct-acting valves and (ii) relay- or pilot-operated types.

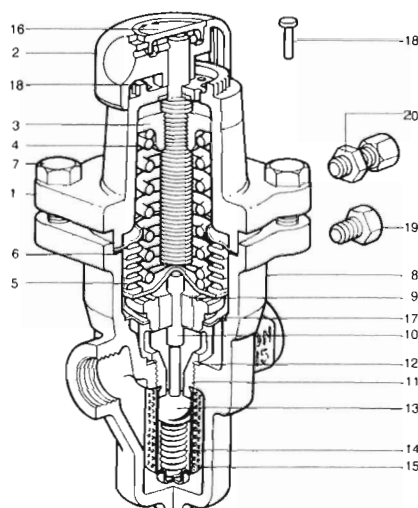
An example of a direct-acting pressure reducing valve for steam, compressed air and other gases is shown in Figure 1. The valve is designed for point-of-use installations. On start-up, the upstream pressure, aided by a return spring, holds the valve head against the seat in the closed position. Downstream pressure is set by rotating a handwheel in a clockwise direction which compresses the control spring and extends the bellows. This downward movement is transmitted via a push rod which causes the main valve to open. Liquid then passes through the open valve into the downstream pipework and also surrounds the bellows. As downstream pressure increases, it acts through the bellows to counteract the spring force and closes the main valve when the set pressure is reached. The main valve modulates to give constant downstream pressure.

Materials used for the bellows include phosphor bronze and stainless steel with nitrile main valve. Other diaphragms used include rubber, synthetic rubbers, stainless steel and phosphor bronze materials.

The direct-acting reducing valve shown in Figure 2 is designed for use with liquids and incorporates a balanced piston design providing accurate control of pressure under stable load conditions. The valve is installed in a horizontal pipeline. Typical applications include laundry equipment and reducing pressure at the point of use on Injection Moulding machines.

Pressure reduction in water systems aids both the efficient design of the piping network and protects consumers from excessive noise from high velocity within buildings, high-pressure discharge at taps and other outlets, and climbing overnight pressures when the distribution network is lightly loaded. The ability to control water-entry pressures ensures a balanced distribution network and also limits the maximum supply volume and so reduces water waste.

Cost-effective steam distribution depends on keeping pipe sizes to the minimum and having the highest acceptable distribution pressure between the boilerhouse and the areas of steam usage, then dropping pressure at the working area to the levels for the highest heat transfer, efficiency and safety.



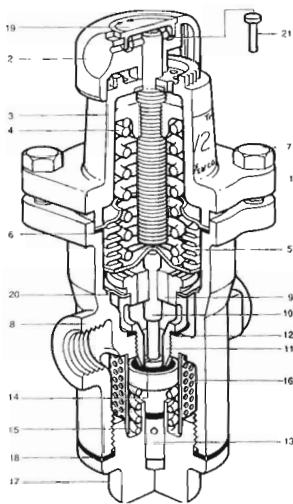
No. Part	Material
1 Spring Housing	Aluminium — Epoxy coated LM 24
2 Adjustment Hand Wheel	Mineral Reinforced Nylon
3 Top Spring Plate	Cast Iron DIN 1691 GG 20
4 Pressure Adjustment Spring	Silicon Chrome BS 2803 685 A55 Spring Steel Range 2
5 Bellows Assembly Option	Stainless Steel 316Ti/316L Phosphor Bronze/Brass
6 Bellows Assembly Gasket	Stainless Steel Reinforced Exfoliated Graphite
7 Spring Housing Bolts	Steel — Zinc plated BS 3692 Gr 8.8 M8 x 25mm
8 Body	Screwed Flanged SG Iron SG Iron DIN 1693 GGG 40.3 DIN 1693 GGG 40.3
9 Guide Bush	Graphite filled PTFE
10 Pushrod	Stainless Steel ASTM A276 316L
11 Valve Seat	Stainless Steel BS 970 431 S29
12 Valve Seat Gasket	Stainless Steel BS 1449 316 S11
13 Valve	Stainless Steel AISI 426
14 Valve Return Spring	Stainless Steel BS 20056 316 S42
15 Strainer Screen	Stainless Steel BS 1449 316 SH
16 Spring Range Identification Disc	Polypropylene
17 Bulkhead Plate	Stainless Steel 316 L
18 Tamperproof Pin	Mild Steel - Copper Plated
19 Blanking Plug	Stainless Steel BS970 431 S29
20 Compression Fitting	Brass

Figure 1. Direct-acting pressure reducing valve.

A relay- or pilot-operated reducing valve for steam services is shown in Figure 3. It comprises:

- The valve body, which contains the main valve and seat piston assembly.
- The control head, which houses the pilot valve assembly with its associated diaphragm and main adjusting spring.

This type of valve works by balancing the downstream pressure against a control spring. This modulates a small valve plug over a seat (the pilot). The flow through this seat is directed in turn to the main valve diaphragm (phosphor bronze or stainless steel), where it modulates the main valve.



Materials

No.	Part	Material
1	Spring housing	Aluminium - Epoxy coated LM 24
2	Adjustment Hand Wheel	Plastic — Polypropylene
3	Top Spring Plate	Cast Iron DIN 1691 GG 20
4	Pressure Adjustment Spring	Silicon Chrome BS 2803 685 A55 Spring Steel
5	Bellows Assembly	Phosphor bronze/brass BS2872 CZ 122 (Stainless Steel optional 316Ti/316L)
6	Bellows Assembly Gasket	Reinforced Exfoliated Graphite
7	Spring Housing Bolts	Steel — Zinc plated BS 3692 Gr 8.8 M8 x 25mm
8	Body	Gunmetal BS1400 LG2
9	Guide Bush	Graphite filled PTFE
10	Pushrod	Stainless Steel ASTM A276 316L
11	Valve Seat	Stainless Steel BS 970 431 S29
12	Valve Seat Gasket	Stainless Steel BS 1449 316 S11
13	Piston	Stainless Steel BS970 431 S29
14	Valve Head	Nitrile Rubber
15	Piston Return Spring	Stainless Steel BS2056 Gr302 S26
16	Strainer Screen	Stainless Steel BS 1449 304 S16
17	Cap	Brass BS 2872 CZ 122
18	Cap Gasket	Reinforced Exfoliated Graphite
19	Spring Range ID Plate	Polypropylene
20	Bulkhead Plate	Stainless Steel 316 L
21	Tamperproof Pin	Mild Steel - Copper Plated

Figure 2. Direct-acting pressure reducing valve for liquids.

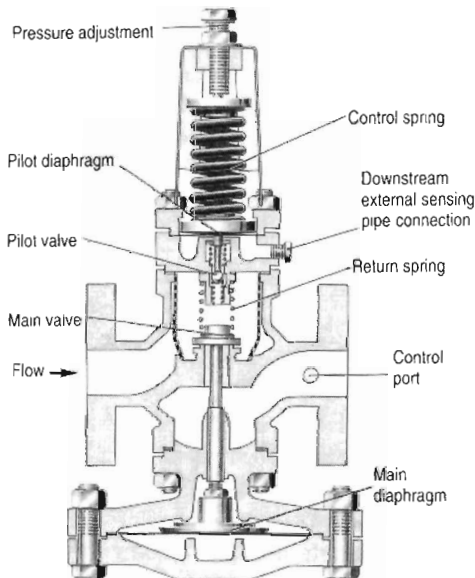


Figure 3. Pilot-operated pressure reducing valve.

Under stable load conditions, the pressure under the pilot diaphragm balances the force set on the adjustment spring. This settles the pilot valve, allowing a constant flow across the main diaphragm. This ensures that the main valve is also settled to give a stable downstream pressure. When the downstream pressure rises, the pilot valve closes and pressure is released from the main valve diaphragm through the control orifice, to close the main valve.

Any variations in load or pressure will immediately be sensed on the pilot diaphragm which will act to adjust the position of the main valve, ensuring a constant downstream pressure. In order to achieve the most stable operating conditions an external pressure sensing pipe is used. This becomes more important as the valve is used near its maximum capacity or under critical flow conditions.

A solenoid will provide for remote on/off control and a fully adjustable set point is possible using an air-driven pilot. The set point can then be adjusted via a compressed air regulator situated away from the valve. For example, the valve may be high up in a pipeline but adjustment can be made from an air regulator at ground level.

The characteristics of both pilot-operated and direct-operated reducing valves are shown in Figure 4. Both curves are shown for 25 mm (1 in) valves reducing from 10 to 3.5 bar (150 to 509 lbf/in²).

It should be noted that in the case of the direct-acting valves (including those with balance pistons) the outlet pressure falls as the flow through the valve increases. Thus if the valve is set at a no-flow setting of 3.5 bar (50 lbf/in²), the outlet pressure falls by about 0.35 to 3.15 bar (5 to 45 lbf/in²) when

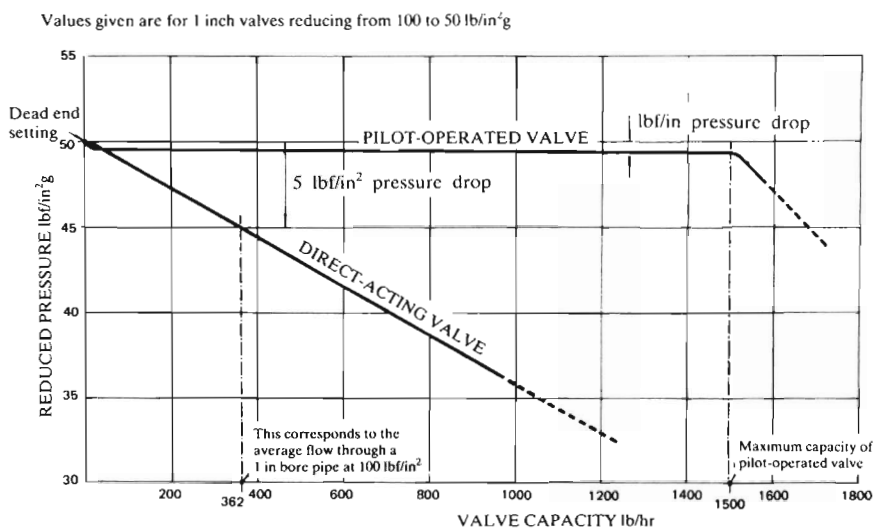


Figure 4. Self-acting reducing valves.

passing an average flow for this type of valve. The direct-acting valve is usually made equal in size to the inlet pipe.

In the case of the pilot-operated valve it will be seen that apart from an initial pressure drop of 0.035 bar (2 lbf/in²) from the dead-end setting of 3.5 bar (50 lbf/in²) the outlet pressure remains constant until maximum-rated capacity is reached.

It should also be noted that, in the examples shown, the 25 mm (1 in) pilot-operated valve is capable of passing a flow of more than four times that of the direct-acting valve, and with only 0.035 bar (2 lbf/in²) pressure drop as compared with a drop of 0.35 bar (5 lbf/in²).

Applications of reducing valves

Reducing valves are used for reducing one pressure to another, control being via throttling of the fluid through the valve and its seat. Reducing valves should never be deliberately oversized as if the valve is too big then the lift of the valve will be small and wire drawing or erosion of the valve and seat can result. Additionally, small variations in valve opening cause large changes in flow which at small flow demands can lead to pulsating pressure being generated in the downstream flow.

The following notes designate the main fields of application of self-acting reducing valves.

Air or gases

This application includes all compressed air systems for use with power tools, pneumatic control systems, etc., and control valves for the storage and distribution of industrial gases, etc.

Both direct-acting and pilot-operated reducing valves may be used for these duties and are selected according to the accuracy of control required and whether or not the valves are intended to give a dead tight shut-off under no-flow conditions.

Water

Reducing valves are extensively used in industrial and domestic water distribution, fire protection systems and the limitation of water pressures in high buildings, etc. Direct-acting valves with piston valves are generally used for these duties. As a general rule, reducing valves are used mainly as pressure limiting devices in water-distribution systems.

Because of high peak demands at times of heavy industrial usage, water authorities usually have great difficulty in maintaining pressures in the systems, although high pressures are usually available at the source of distribution, such as at reservoirs or main pumping stations. Very large pressure drops are

experienced in the system during high demands and as a result there is a tendency for the pressure to be below normal at the point of usage.

However, when the total demand in the system drops, much higher pressures are experienced in the distribution system and these are frequently in excess of the normal pressure ratings for the equipment being used. This can give rise to burst mains or excessively high discharge rates from domestic fittings, such as water closets, wash basins, etc. or storage tanks.

It is therefore common practice to fit a reducing valve in the line which, under high flow conditions, normally operates in the wide open position and presents only a nominal resistance to flow (such as would be experienced with an ordinary globe stop valve). However, at periods of low demand and high pressure, the reducing valve becomes effective and reduces the pressure in the downstream mains to an acceptable limit. It is important in such applications that the outlet pressure is not affected by inlet pressure variations and for this reason the direct-acting valves with piston balance are admirably suited to this application.

Other liquids

In this field, reducing valves are used for such applications as: controlling ram pressures on hydraulic presses; bearing lubrication systems in rolling mills and heavy industrial equipment; and for pressure control in fuel-oil systems. Again valves are normally used for these applications. In many applications the flow is relatively constant and the outlet pressure from the reducing valve therefore remains constant.

In fuel-oil systems the flow variations are normally of the order of 50 to 100%, in which case the outlet pressure variation would probably be of the order of 0.14 to 0.21 bar (2 to 3 lbf/in²), depending on the size of valve used. The variation would be in the order of 0.35 bar (5 lbf/in²) between full and no-flow conditions.

Steam

This particular category covers by far the majority of reducing valve applications and in general there are two broad sections.

Power

Reducing valves are only occasionally used on power installations involving steam, i.e. direct steam supply, steam engines and turbines, etc. In these cases the general principles of application still apply, although special problems do sometimes arise in the case of reciprocating machinery which may give rise to pulsations in the pipework system, and these can be amplified by the reducing valve itself. This is normally overcome by providing adequate pipe volume both upstream and downstream of the reducing valve to act as a 'steam accumulator'.

Process

With saturated steam, temperatures and pressures are strictly related and, because of this, it is frequently found convenient to control temperature by controlling the steam pressure. Applications in the process field includes space heating, kitchen equipment, sterilizing equipment, curing processes in the rubber and plastics industries, etc., industrial cooking equipment, etc. In fact, anywhere steam is used as a heat-transfer medium, reducing valves will invariably be installed.

In general, only low pressure steam, usually below 3.5 bar (50 lbf/in²), is used for process purposes. At such low pressures the latent heat content of the steam is relatively high and is easily transferred from the steam to the product being processed.

Size of pipes and fittings

The inlet and outlet pipes should be sized to suit the maximum steam demands of the system, e.g. see Table 1. Pipe sizes should always be determined in terms of pressure drop and not by such rules as arbitrary steam velocities.

Correct sizing of pipework and fittings associated with all valves is extremely important in order to obtain the best possible operation. Specifically:

- (i) Strainers should always be equal in size to the inlet pipe.
- (ii) When globe valves are used as inlet and outlet stop valves, these should also be equal in size to the respective pipe, into which they are fitted.
- (iii) When parallel slide valves are used as stop valves, these can be fitted equal in size to the reducing valve for reduced pressures between 30 and 70% of the inlet pressure. They should be equal in size to the respective pipe when the pressure difference between the inlet and outlet is 2 bar (30 lbf/in²) or less. When they are connected directly to the reducing valve the length of distance pieces between valve and fitting should not exceed three pipe diameters.
- (iv) In order to provide a streamlined flow at the approach to the reducing valve, a straight length of pipe equal to 10 pipe diameters should be provided between the fitting and the reducing valve (this does not apply to parallel slide valves). Typical reducing valve layouts are shown in Figures 5 and 6.

Reducing valves in parallel

As already mentioned, two reducing valves in parallel should be considered when the minimum flow through the system is less than 10% of the maximum capacity of a single reducing valve, or when the valves are expected to work for long periods on 'no-flow' or 'dead-end' conditions, or working in partially completed plant systems. A typical layout is shown in Figure 7.

Table 1. Steam pipe capacities (lb/hr dry saturated)

Pressure lb/in ² bar	Pipe size in inches												
	1/2	3/4	1	1 1/4	1 1/2	2	2 1/2	3	4	5	6	8	10
5 <i>0.35</i>	12 <i>0.4</i>	32 <i>0.4</i>	63 <i>0.4</i>	106 <i>0.4</i>	163 <i>0.4</i>	320 <i>0.4</i>	536 <i>0.4</i>	813 <i>0.4</i>	1560 <i>0.4</i>	2550 <i>0.4</i>	3820 <i>0.3</i>	6180 <i>0.3</i>	100,000 <i>0.3</i>
10 <i>0.09</i>	15 <i>0.5</i>	40 <i>0.5</i>	79 <i>0.5</i>	113 <i>0.5</i>	206 <i>0.5</i>	404 <i>0.5</i>	676 <i>0.5</i>	1055 <i>0.5</i>	1962 <i>0.5</i>	3220 <i>0.5</i>	4810 <i>0.5</i>	8000 <i>0.4</i>	12,950 <i>0.4</i>
20 <i>1.38</i>	18 <i>0.6</i>	54 <i>0.7</i>	107 <i>0.7</i>	182 <i>0.7</i>	300 <i>0.8</i>	586 <i>0.8</i>	980 <i>0.8</i>	1400 <i>0.7</i>	2680 <i>0.7</i>	4390 <i>0.7</i>	6550 <i>0.7</i>	11,300 <i>0.6</i>	15,650 <i>0.5</i>
30 <i>2.07</i>	24 <i>0.8</i>	69 <i>0.9</i>	137 <i>0.9</i>	245 <i>0.9</i>	377 <i>1.0</i>	740 <i>1.0</i>	1240 <i>1.0</i>	1880 <i>0.9</i>	2600 <i>0.9</i>	5900 <i>0.9</i>	8810 <i>0.9</i>	14,700 <i>0.8</i>	22,300 <i>0.7</i>
50 <i>3.45</i>	35 <i>1.2</i>	98 <i>1.3</i>	203 <i>1.4</i>	357 <i>1.5</i>	582 <i>1.6</i>	1140 <i>1.6</i>	1910 <i>1.6</i>	2640 <i>1.4</i>	4880 <i>1.3</i>	8000 <i>1.3</i>	12,000 <i>1.3</i>	21,450 <i>1.2</i>	33,400 <i>1.1</i>
75 <i>5.17</i>	47 <i>1.6</i>	136 <i>1.8</i>	284 <i>2.0</i>	495 <i>2.1</i>	780 <i>2.2</i>	1493 <i>2.1</i>	2500 <i>2.1</i>	3800 <i>2.1</i>	7100 <i>2.0</i>	11,600 <i>2.0</i>	16,900 <i>1.9</i>	30,600 <i>1.8</i>	48,400 <i>1.7</i>
90 <i>6.21</i>	53 <i>1.8</i>	157 <i>2.1</i>	332 <i>2.4</i>	581 <i>2.5</i>	896 <i>2.5</i>	1755 <i>2.5</i>	2940 <i>2.5</i>	4460 <i>2.5</i>	8280 <i>2.4</i>	13,570 <i>2.4</i>	19,450 <i>2.2</i>	35,400 <i>2.1</i>	54,500 <i>1.9</i>
100 <i>6.90</i>	59 <i>2.0</i>	173 <i>2.3</i>	362 <i>2.6</i>	615 <i>2.6</i>	984 <i>2.8</i>	1928 <i>2.8</i>	3220 <i>2.8</i>	4840 <i>2.7</i>	9040 <i>2.6</i>	14,820 <i>2.6</i>	21,800 <i>2.5</i>	38,900 <i>2.3</i>	61,500 <i>2.2</i>
120 <i>8.27</i>	68 <i>2.3</i>	204 <i>2.8</i>	431 <i>3.2</i>	755 <i>3.4</i>	1182 <i>3.5</i>	2320 <i>3.5</i>	3900 <i>3.5</i>	5710 <i>3.3</i>	10,920 <i>3.3</i>	17,910 <i>3.3</i>	26,000 <i>3.1</i>	46,900 <i>2.9</i>	73,000 <i>2.7</i>
150 <i>10.34</i>	83 <i>2.8</i>	252 <i>3.5</i>	539 <i>4.1</i>	945 <i>4.4</i>	1420 <i>4.5</i>	2900 <i>4.4</i>	4770 <i>4.2</i>	7100 <i>4.2</i>	13,600 <i>4.2</i>	22,300 <i>4.2</i>	32,500 <i>4.0</i>	59,100 <i>3.8</i>	93,000 <i>3.6</i>
180 <i>12.41</i>	96 <i>3.3</i>	296 <i>4.2</i>	642 <i>5.0</i>	1131 <i>5.3</i>	1748 <i>5.4</i>	3440 <i>5.5</i>	5890 <i>5.7</i>	8750 <i>5.5</i>	16,680 <i>5.4</i>	27,300 <i>5.4</i>	40,000 <i>5.2</i>	71,500 <i>4.8</i>	112,000 <i>4.5</i>
200 <i>13.79</i>	107 <i>3.7</i>	324 <i>4.6</i>	708 <i>5.6</i>	1238 <i>5.9</i>	1942 <i>6.1</i>	3880 <i>6.3</i>	6530 <i>6.4</i>	9750 <i>6.4</i>	18,880 <i>6.3</i>	31,000 <i>6.3</i>	44,300 <i>5.8</i>	80,000 <i>5.5</i>	126,000 <i>5.2</i>
220 <i>15.17</i>	116 <i>4.0</i>	354 <i>5.0</i>	770 <i>6.0</i>	1350 <i>6.4</i>	2120 <i>6.6</i>	4200 <i>6.8</i>	7150 <i>7.0</i>	10,850 <i>7.0</i>	20,800 <i>7.0</i>	34,000 <i>7.0</i>	49,000 <i>6.5</i>	90,000 <i>6.3</i>	141,000 <i>5.9</i>
250 <i>17.24</i>	133 <i>4.5</i>	408 <i>5.7</i>	871 <i>6.7</i>	1525 <i>7.1</i>	2385 <i>7.3</i>	4760 <i>7.6</i>	8100 <i>7.8</i>	12,360 <i>8.0</i>	23,800 <i>8.0</i>	39,000 <i>8.0</i>	57,000 <i>7.6</i>	105,000 <i>7.8</i>	168,000 <i>7.0</i>
300 <i>20.69</i>	157 <i>5.3</i>	496 <i>7.2</i>	1025 <i>7.8</i>	1798 <i>8.3</i>	2800 <i>8.5</i>	5160 <i>8.9</i>	9550 <i>9.2</i>	14,760 <i>9.6</i>	28,400 <i>9.6</i>	46,500 <i>9.6</i>	68,900 <i>9.4</i>	125,900 <i>9.1</i>	202,000 <i>8.9</i>

Note: Figures in *italic* show pressure drops (lb/in²) for equivalent lengths equal to 360 pipe diameters.

When using this table, allowance should be made for the effects of bends and fittings in the pipeline.

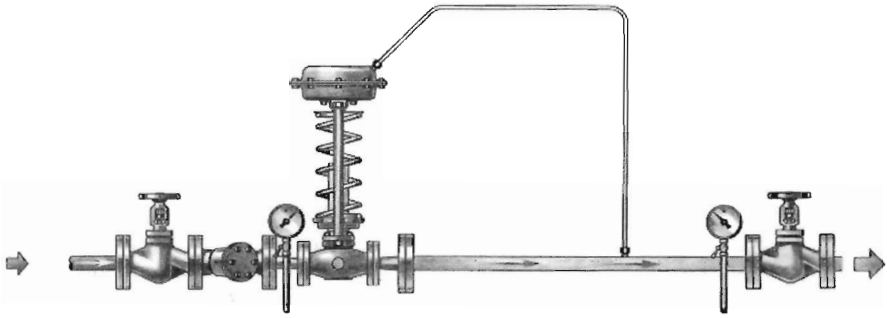


Figure 5. Control of water distribution.

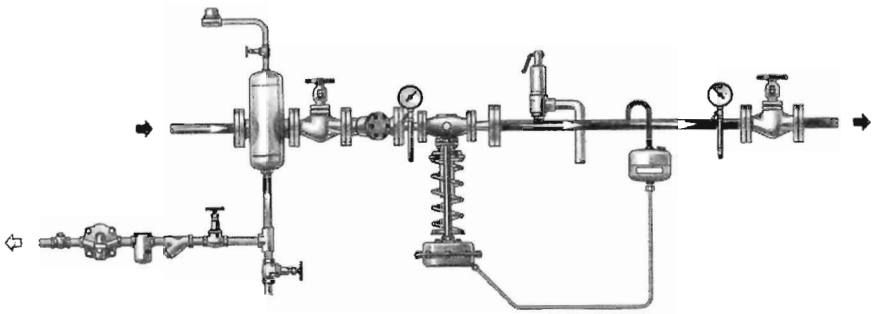


Figure 6. Control of steam distribution.

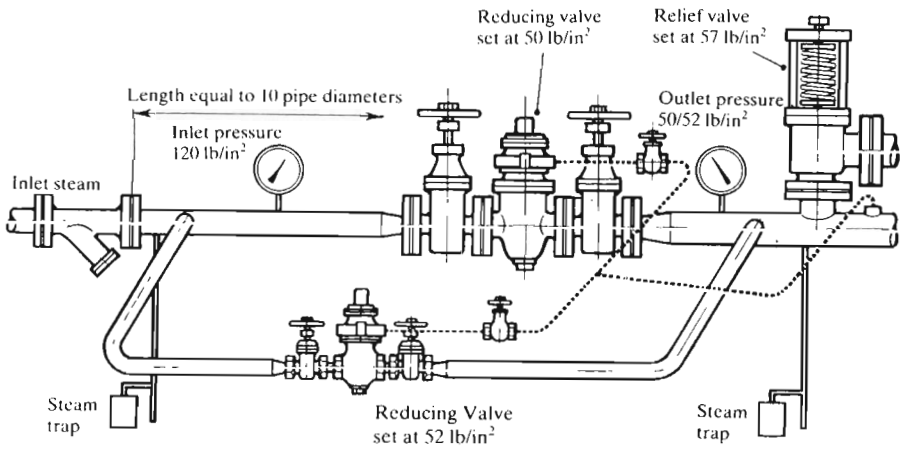


Figure 7. Parallel arrangement of reducing valves.

In order that the valves can deal effectively with minimum capacity variations of less than 10%, two unequally sized reducing valves having a maximum capacity equal to the required capacity should be connected in parallel with the outlet pressure of the smaller valve set 0.14 to 0.21 bar (2 to 3 lbf/in²) higher than that of the large valve. In this way the larger valve would shut at low demands leaving the smaller valve to handle the low flows. As the demand increases the larger valve will open automatically as the reduced pressure falls, and share the load with the small valve. By this method capacity ratios of up to 100:1 can be obtained.

Superheated steam

Superheated steam is less dense than saturated steam and, therefore, for the same pressure drop the reducing valve will have slightly smaller capacity. The reduction in capacity is dependent on the amount of superheat. Capacity figures quoted in manufacturers' catalogues are normally for dry saturated steam. When steam is superheated above 28°C (50°F) before it enters the reducing valve, dry steam capacities should be multiplied by the following factors:

56 to 83°C (100 to 150°F) of superheat—0.89

83 to 111°C (150 to 200°F) of superheat—0.86

111 to 167°C (200 to 300°F) of superheat—0.82

Steam traps

Whenever possible, pilot-operated reducing valves should be sited at some point in the pipeline where they cannot become flooded with condensate during periods of low flow or prolonged shutdown. If this is not possible then steam traps (and, if possible, dirt pockets) should be fitted to both the inlet and outlet pipework to remove any condensate which may accumulate in the vicinity of the reducing valve.

Condensate can be trapped between the piston and pilot valve when the steam flow is resumed and this prevents the main valve from closing as the reduced pressure may continue to rise above the setting and eventually cause the relief valve to blow.

The thermodynamic steam trap shown in Figure 8 is suitable for use in condensate removal from high pressure steam mains and for turbine casing drainage.

On start up, incoming pressure raises the disc and cooled condensate plus air is immediately discharged (A). Hot condensate flowing through the trap releases flash steam. High velocity creates a low pressure area under the disc and draws it towards the seat (B). At the same time there is a pressure build-up of flash steam in the chamber above the disc, which forces it down against the

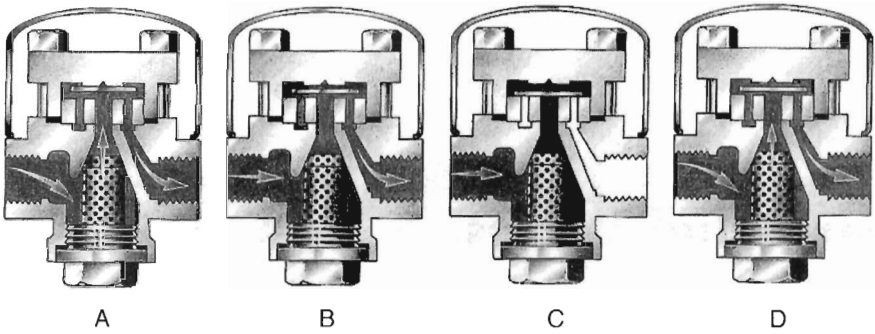


Figure 8. Thermodynamic steam trap—operating sequence.

pressure of the incoming condensate until it seats on the inner ring and closes the inlet. The disc also seats on the outer ring and traps pressure in the chamber (C). Pressure in the chamber is decreased by condensation of the flash steam and the disc is raised by the incoming pressure. The cycle is then repeated (D).

Fitting of balance pipes

It is strongly recommended that a balance pipe should be fitted when the reduced pressure is 10% or less of the inlet pressure. The purpose of this pipe is to improve the performance of the reducing valve when working under difficult downstream conditions. It will also help to counteract any pressure drops in downstream pipework caused by undersized pipe fittings, etc., providing they are not excessive.

A stop valve should be fitted in the balance pipe to allow complete isolation of the reducing valve from the steam flow (particularly when a bypass line is fitted).

The balance pipe should be arranged to fall to allow it to drain into the downstream pipe. The tapping into the downstream pipe should be made at a point where smooth flow occurs preferably downstream of the relief valve. The downstream pressure gauge should be fitted as near to this point as possible.

Air Relief Valves

Air or gas trapped in a pipeline carrying a liquid can cause problems, e.g. reduce the effective flow, aggravate the effects of surge, and cause pump cavitation. Possible causes of air/gas entrainment are:

- (i) The pipeline was fully charged with air/gas when empty.
- (ii) Air is entrained at pump suction.
- (iii) Air is drawn in through faulty joints or glands.
- (iv) Air/gas is trapped in pockets during pipeline filling.
- (v) Air/gas in solution is released due to changes in pressure and temperature.

The problem of dealing with air/gas entrainment is not usually a demanding one. Entrained air/gas will tend to collect at high points in the system. It can then be removed by introducing air release valves at these points. These may be simple, manually-operated valves (bleed valves), or fully automatic. In the latter case the valve should perform the following functions:

- (a) release of air/gas accumulating in pipeline during normal pressurised operation, to prevent restriction to fluid flow
- (b) retention of the fluid in the pipeline without loss under all operating conditions
- (c) release of air/gas during pipeline filling at a volume rate sufficient to prevent back pressure restricting the filling rate
- (d) admission of air to the pipeline during emptying at a rate sufficient to prevent excessive vacuum pressure in the pipe.

Single orifice valves (Figure 1) are capable of performing functions (a) and (b). They are normally used where only relatively small volumes of air/gas are to be released, or where it is desirable to provide additional ventilation at operating pressures. Dual-orifice valves are capable of performing all four functions. They can normally provide complete protection against air/gas entrainment under all system-operating conditions.

The type of fluid product being handled also affects the design requirements of the air release valve (especially automatic valves). With sewage or industrial

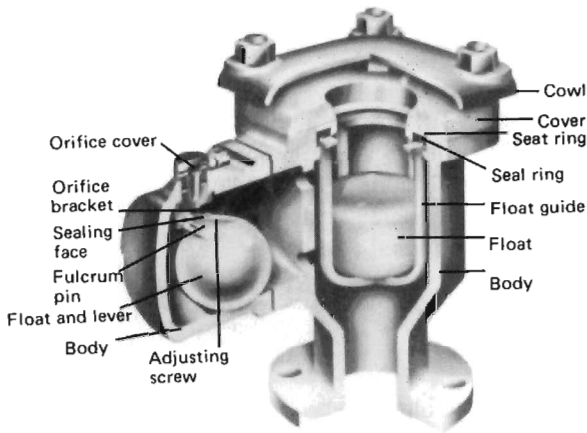


Figure 1. Single-orifice air relief valve.

effluent, for example, the solids content may block the release passage(s) periodically, causing unreliable operation. This can be overcome by using large-volume auxiliary float chambers to contain the fluid under all operating conditions so that it can never come into contact with the air valve elements.

An example of a dual-orifice air valve suitable for water systems is shown in Figure 2.

The valve combines small and large orifices. The small-orifice valve comprises a composite float and lever assembly sealing off a small-orifice vent. When the

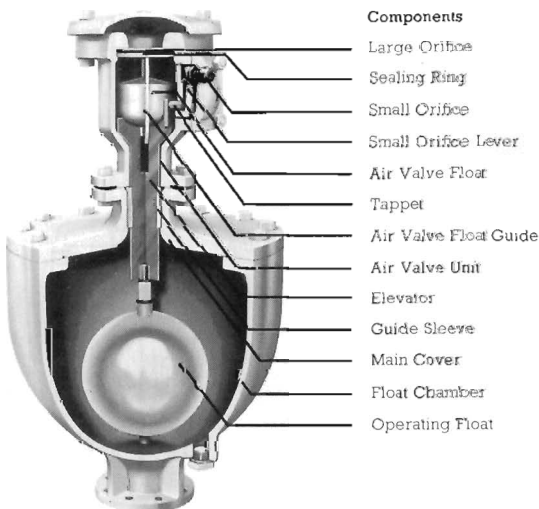


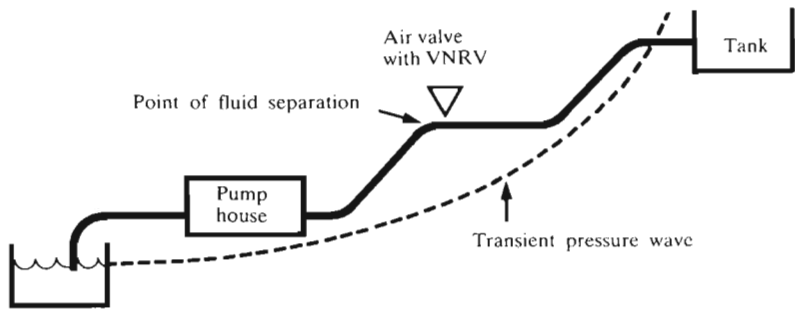
Figure 2. Dual-orifice air relief valve.



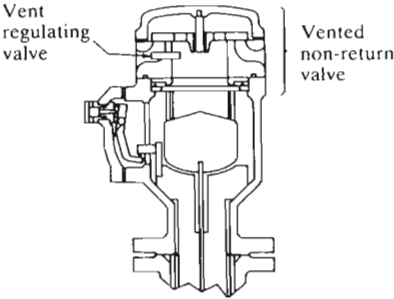
Underpressure air release valve.



Dual-orifice air valve.



Example of patented air relief valve with vented non-return valve for performing additional function of surge suppression.



Air relief valve.

float chamber is filling with water, the orifice is closed initially by the float working through a lever ratio of 5:1.

When the chamber is filled with water under pressure, the orifice is held closed by the combined upthrust of the float and the differential pressure over the orifice area. On air accumulated in the system entering the chamber under working pressure, the water level in the chamber is depressed until it reaches a point when the weight of the float is sufficient to uncover the orifice and exhaust air. Air is expelled until the water level rises again and causes the float to close the orifice.

The large-orifice valve consists of a float sealing off a large-orifice vent to the atmosphere. The float is held at a predetermined height in its casing by a ribbed cage which also guides the float onto the seat. During the pipeline filling or emptying, the 'aerokinetic' feature holds the float off the seat and keeps it completely stable under all air outflow or inflow conditions. The valve cannot close prematurely during outflow. It closes only when water enters the casing and raises the float onto the seat.

An example of the working of dual-orifice air relief valves designed for handling sewage and similar effluent is shown in Figure 3. When the pipeline

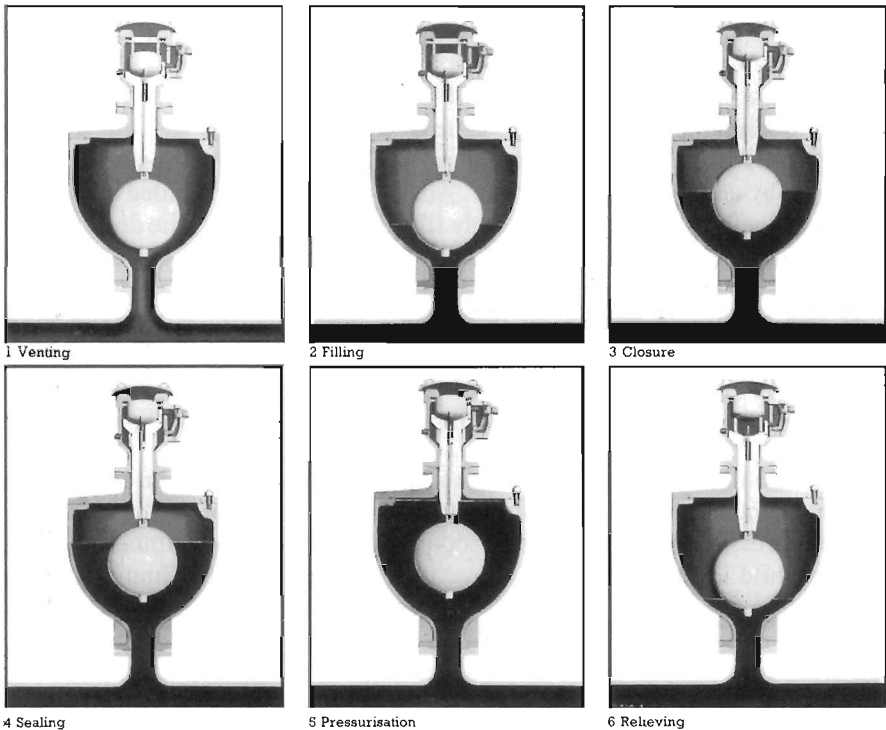
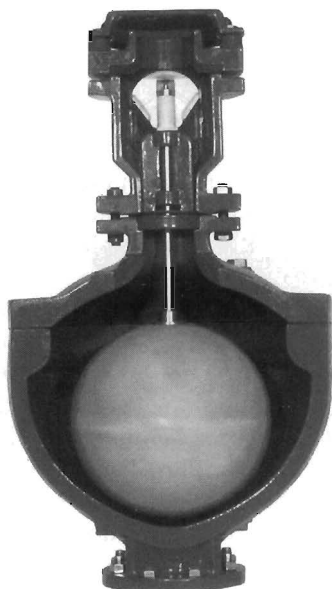


Figure 3. Operational sequence of a dual-orifice air relief valve.



Double-acting sewage air valve.

is empty, the spherical operating float is suspended from the elevator within the main chamber and the cylindrical float element of the air valve is supported by the guide cage. The operating lever of the small-orifice valve is held open by a tappet on the elevator. Air/gas having been inhaled or expelled from the pipelines is able to flow freely through both orifices, the design of the valve being such that the air flow creates a positive down force to hold the cylindrical float element stable within the guide cage.

As the air/gas is exhausted from the pipeline, liquid enters the main chamber and the operating float then rises with the liquid. The elevator, raised by the float, releases the small-orifice valve and engages the base of the cylindrical element, which rises until seated on the rubber face of the large orifice.

At this point air/gas outflow ceases and further inflow of liquid to the main chamber under pipeline pressure compresses the air/gas until maximum working pressure is reached. The proportions of the main chamber are such that the fluid level will not rise above the bottom face of the main chamber cover.

When the pipeline is emptied and pressure falls, the valve main chamber will drain into the pipeline and the operating float, following the liquid level, releases the cylindrical float to allow the large orifice to open. As the pipeline pressure falls to atmospheric, it opens the small-orifice valve. The pipeline is then able to ventilate freely and sub-atmospheric pressure conditions are avoided.

During normal operating conditions, air/gas will be released from the liquid and will collect under pressure in the main chamber, depressing the liquid level. The operating float falls with the liquid but system pressure will hold the cylindrical valve element on the large-orifice seat. As the operating float approaches the limit of its travel, the tappet on the elevator opens the small-orifice valve, releasing the accumulated air/gas under pressure. This in turn allows the liquid level to rise again and the small-orifice valve to close, thus completing a cycle.

Positioning of air relief valves

In systems handling water, air relief valves would normally be placed at all high points, i.e. where a rising section changes to a falling section. In systems handling sewage or industrial effluent, rather more extensive treatment is necessary, as illustrated in Figure 4.

Where the fluid is pumped through the pipeline it is desirable that a dual-orifice valve (valve A) be located just downstream of the pump-delivery valves.

Dual-orifice-type valves are also required at all peak points which are defined relative to the hydraulic gradient and not necessarily to the horizontal. In practice a peak may be considered as any pipe section which slopes up towards the hydraulic gradient or runs parallel to it. In the latter case the

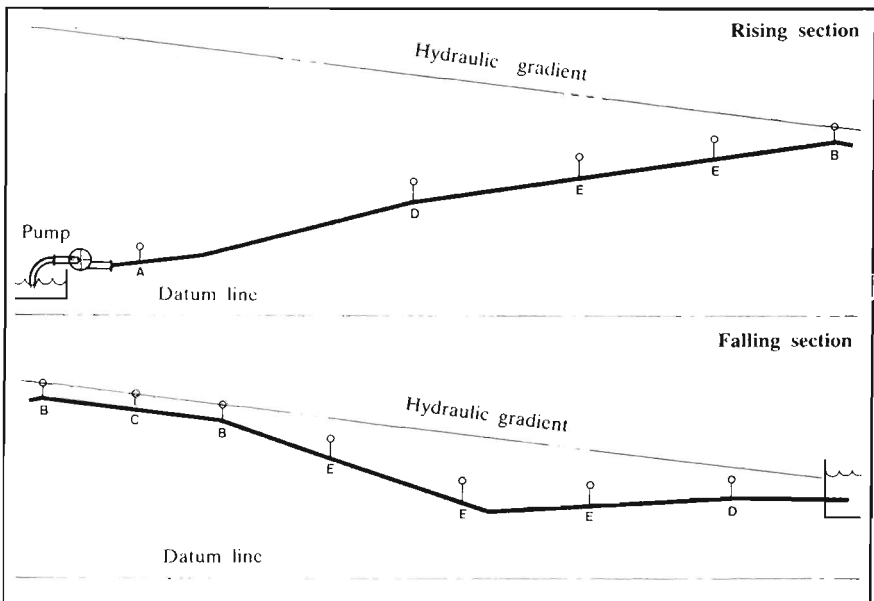


Figure 4. Typical sewage air valve location points.

minimum requirement is a dual air valve at each end of the section (valve B); any additional valves may be of a single-orifice type.

Positions where an increase in down slope occurs will require ventilation by a small orifice valve which should also be installed at points of decrease in up slopes (valve D).

Pipeline sections of uniform profile also require ventilation and dual-orifice units should be installed at about 800 m (2500 ft) intervals on these sections (valves E).

Foot Valves

A foot valve is basically a check valve fitted to the end of a suction pipe leading to a pump. Its purpose is to keep fluid trapped in the suction pipe when the pump stops, thus maintaining a suitable prime for the pump. When the pump restarts, the suction created opens the valve, giving full flow to the pump inlet. (Foot valves are unnecessary on self-priming pumps.)

Foot valves may be of a simple flap-type, or more usually lift-check or ball-check valves. They are commonly combined with an integral strainer. Some examples follow.

Poppet lift foot valve

In the example shown in Figure 1, the poppet assembly consists of a plastic tripod which can be displaced along a bore above the seat valve. The travel of the poppet is controlled by a stop on the end of the poppet legs acting as supports for the return spring shouldered onto a washer. This spring ensures that the valve will work in any position. The main characteristics of this design are low head losses with good sealing provided by a nitrile rubber O-ring.

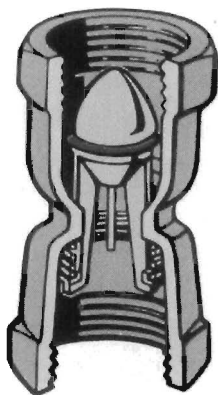


Figure 1. Foot valve with plastic tripod.

Figure 2 shows a design with the tripod in cast iron and with a cast-iron poppet head with streamlined tripod hub. Sealing is provided by a flat gasket shouldered by the poppet head and placed on a collar-type seat. This is a simple and robust design suitable for general applications.



Poppet-type foot valve with strainer.

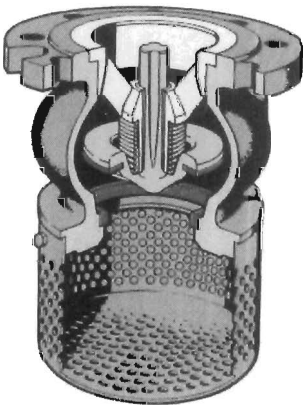


Figure 2. Foot valve with tripod and poppet head in cast iron.

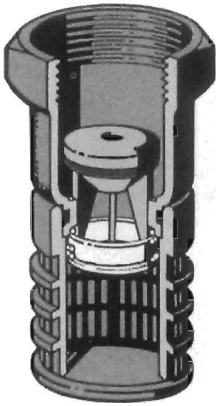


Figure 3. Foot valve with all metal poppet and profiled head.

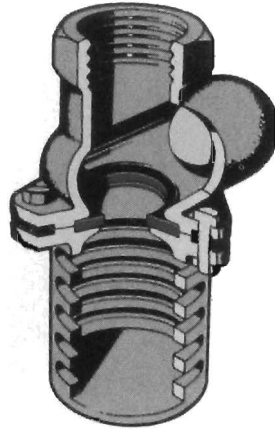


Figure 4. Ball foot valve.

Figure 3 shows a further design where the all-metal poppet with profiled head is guided by three legs and restrained by a downstream stop. Sealing is by a flat seal on a flat bearing surface. Valve travel is limited by the stop. A spring can be added to ensure that the valve will operate in any position.

Ball foot valve

An example of this type is shown in Figure 4. This is a simple ball valve guided by an inclined cylindrical chamber and seating on an O-ring. Note that the ball is displaced laterally along its chamber with inward flow, but it runs down

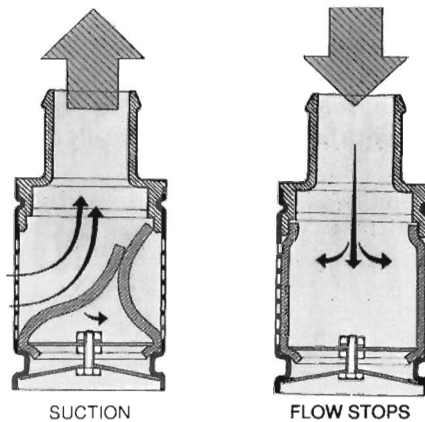


Figure 5. Membrane foot valve.



Figure 6. A selection of membrane foot valves.

the chamber onto its seat when the flow rate decreases. It is particularly suitable for use with contaminated waters or more viscous fluids.

All examples illustrated are of the type with integral strainer.

Membrane foot valves

Membrane foot valves consist of a cylindrical rubber membrane fitted inside a steel strainer. When there is a suction developed at the strainer, the membrane is displaced to allow fluid to flow through the valve. When back-flow conditions exist, the cylindrical membrane closes the apertures in the valve strainer, thus closing the valve, (see Figure 5).

A selection of membrane foot valves is shown in Figure 6. The lever fitted to one valve enables the valve to be drained by physically displacing the membrane when the lever is lifted.

See also the chapter on Check Valves.

SECTION 4

Control and Automation

Valve Actuators

Control Valves

Float Control Valves

Temperature Control Valves

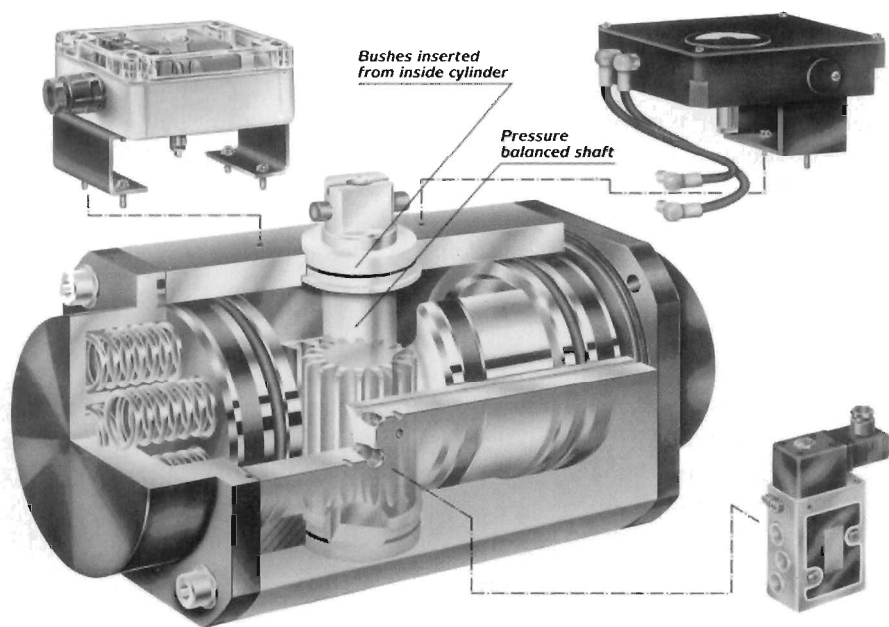
Regulators

Valve Actuators

Numerous types of devices exist for the remote operation of valves. These range from simple gearboxes to highly sophisticated motorised valves with automatic control, programmable logic controllers, microcomputers and field communications networks.

In basic terms, an actuator can be described as 'A device supplying force and motion to the closure member (ball, disc, plug, etc.) of a valve'.

There is a distinction between the actual requirements but, in general, the vast majority of applications are concerned with the opening and closing of



Standard-type pneumatic valve actuator.

valves. Certain systems may call for continuous modulating control which can set limits on the usefulness of both mechanical and energy systems.

In 1992, the world's first non-intrusive 'intelligent' enclosed actuator, that could be commissioned and interrogated without removing electrical covers, was launched.

All actuator settings and diagnostics are made through a sealed indication window using an infra-red setting tool, avoiding the use of penetrating shafts. Solid-state torque and limit measurement is used throughout, eliminating the use of springs, switches or levers.

The separation of the setting tool from the actuator provides a most effective method of security of the settings.

Multi-turn actuators—on-off duty

The general design principle of a multi-turn actuator is to turn a multiple of 360° at its output drive. The resulting rounds-by-stroke relationship can be from 2 to more than 1450. Within this range, position-indicating devices, such as limit switches, and electronic position transducers can be adjusted. Initially, multi-turn actuators fitted to valves transform the cycle movement of the output drive to linear movement according to the demands of the valve. Multi-turn actuators for on-off duty (short-time operation 10–15 min) mostly operate valves with only a few number of opening and closing periods a month.

Part-turn actuators—on-off duty

There is another type of multi-turn actuator for operating ball or butterfly valves or dampers, for example, with a movement of less than 360° for opening and closing.

Normally, the internal gear of the part-turn actuator is designed for a turn of 90° , although 120° and 180° are commonplace.

Thrust actuators—on-off duty

Thrust actuators may be fitted to valves requiring linear movement. They transform the torque of a multi-turn actuator into an axial thrust force by means of an attached thrust unit. Linear actuators are mainly used with globe valves or similar and, when indirectly mounted, may be used to operate butterfly valves, flap valves or dampers.

Multi-turn actuators—modulating duty

The requirements of modulating service (operation of control valves) on actuators are different from the 'normal' on-off duty.

Part-turn actuators—modulating duty

Typically, a part-turn actuator for operating a control valve consists of a combination of a multi-turn actuator with a worm gear, the connection between the two being achieved via the output drive.

It is probably more appropriate to deal with the overall subject of actuators under three main headings:

- (i) main types of valve actuators
- (ii) choice of energy systems
- (iii) electric/electronic controls

Main types of valve actuators

The main types of actuators used are:

- (a) manual operators
- (b) cylinder actuators (pneumatic or hydraulic)
- (c) vane actuators
- (d) electric solenoid
- (e) diaphragm actuators
- (f) electric motor actuators

All have their particular virtues and uses.

Manual operators

Apart from the obvious job of providing a means whereby a person can open or close a valve, the requirements of a manual valve actuator may encompass any or all of the following:

Convert motion from linear to rotary

Valves of the gate, globe, diaphragm and pinch type utilise linear motion of the obturating members to achieve a seal. Handwheel operation implies rotary motion, and conversion is generally by nut-and-screw, which may be part of the valve or of a separate manual actuator.

Withstand thrust

When the threaded nut forms part of the manual actuator, it will have to withstand the operating thrust developed and incorporate suitable thrust bearings.

Lock the valve in position between operations

For valves operated by linear screw thread, position locking is achieved by use of an irreversible, (i.e. low-efficiency) thread. Butterfly valves must also be restrained against self-operation as a result of dynamic flow, and again this is customarily ensured by using irreversible, low-efficiency, gearing such as worm and wheel between valve stem and handwheel.

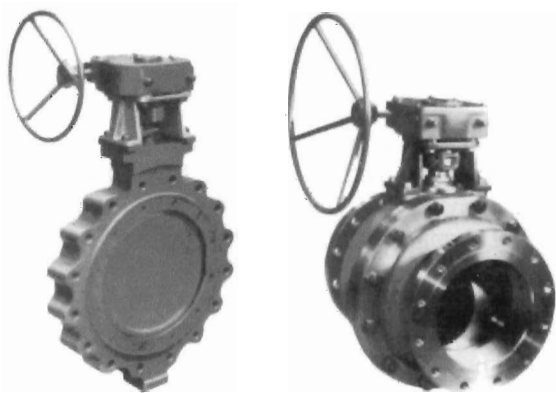
In each case the efficiency must be low enough so that the valve does not move from an intermediate position if pressure is applied to the stationary valve, and it will also stay in position if manually-operated with flow present. There have been many occasions when valve slamming has occurred under high-flow conditions with a gear ratio which was thought to be irreversible, but proved not to be when the handwheel was started under heavy-flow conditions and the valve took over.

This dynamic effect is less of a problem with plug and ball valves which have a higher ratio of static friction to dynamic torque and, in any case, are not suited in standard form to flow regulation.

Whatever manual means is adopted for the operation of valves, the main constraint is always the human muscle power available. It is difficult for a person to exert more than about 75 W (1/10 hp) continuously for any length of time by hand.

Size and type of valve, line pressure and other factors will determine the power required. For larger valves, this will nearly always mean the introduction of an intermediate gearbox with a handwheel capable of operating the valve comfortably with a human's strength—i.e. with a rim pull below 27 kg (60 lb).

The average male can exert up to three times this force momentarily, by pulling his weight on a handwheel, so that the extra force required to seat or unseat a valve is not a problem if the gearing is adequate. However, the



Manual gear actuators.

fundamental principle of gearing to reduce input torque inevitably means that the number of turns required of the handwheel increases. For large valves and pressures, the number of turns is such that manual operating time may be measured in hours rather than minutes.

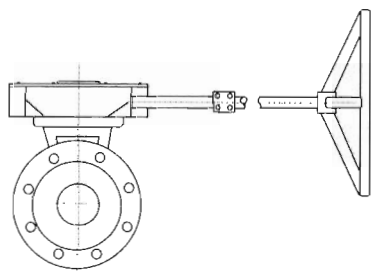
There is no way, therefore, that emergency manual operation can be quick. The alternative philosophy is to accept that large valves are difficult to operate and require at least two people, or a team in an emergency, operating a larger handwheel with far fewer turns. There is no ready solution to this dilemma other than power operation.

Another reason for utilising reduction gearing is to reduce the torque to be transmitted via shafts and couplings to a remote operating point, e.g. spur gearing of a gate valve to a pedestal, say, on a floor above.

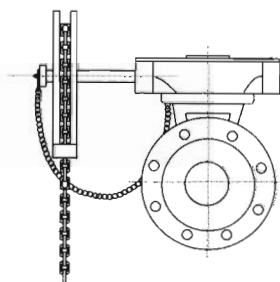
A side handwheel is generally more convenient than one with a vertical shaft. Bevel gear actuators are therefore widely used to reduce the operating torque of gate valves and bring their handwheel to a particular side for access.

The use of worm gearboxes and equivalent nut-and-screw scotch yoke actuators for quarter-turn valves automatically turns the drive through a right-angle, which may therefore be oriented in the right direction. Additional

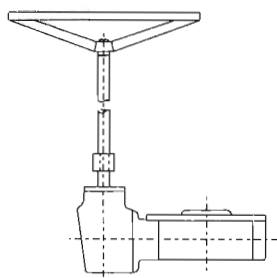
HANDWHEEL EXTENSION



CHAIN SPROCKET



RIGHT-ANGLE DRIVE



Manual operators.